

# AMIRIS Germany2027 - Progress Report

*Everything done since Energy Brainpool's real 2027 input data was supplied, through this week's correlation investigation*

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## Executive Summary

Starting from Energy Brainpool's real 2027 forecast data (installed capacities, demand components, and fuel/carbon prices), a full AMIRIS Germany2027 scenario was built, documented, refined, and then validated against Brainpool's own real 2027 hourly electricity price forecast - the first genuine AMIRIS-vs-Brainpool comparison of this thesis. The first comparison showed AMIRIS's simulated prices looking wildly higher than Brainpool's (by 3-7x on average). Rather than accept that gap, it was diagnosed down to a specific, fixable cause - and after a fix and a calibration exercise, AMIRIS's price forecasts now land close to Brainpool's own numbers.

| Stage                                                         | Headline result                                                                                                                                                                                                                                                           |
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| <b>Initial build (V1, no import)</b>                          | Physically ran and cleared the market, but shortage pricing hit 15.9% of the year - the model was clearly missing something.                                                                                                                                              |
| <b>Refined demand model (V2)</b>                              | Cut shortage hours to 7.5% and mean price by 45% just by modelling demand more realistically.                                                                                                                                                                             |
| <b>Added cross-border imports</b>                             | Helped further (V2: shortage down to 5.4%) but the import model itself was flawed - only 7-12% of the intended import volume ever actually cleared.                                                                                                                       |
| <b>Compared against Brainpool's real 2027 price</b>           | AMIRIS looked 3-7x too expensive on average - but on the 92-95% of hours without a supply shortage, AMIRIS's price level was already close to Brainpool's.                                                                                                                |
| <b>Diagnosed the gap</b>                                      | 64% of AMIRIS's shortage hours had literally zero import available that hour - the import timing was copied from an unrelated year (2023) with no connection to this build's own scarcity pattern.                                                                        |
| <b>Fixed and calibrated the import model</b>                  | V1 shortage hours: 11.8% -> 0.83%, mean price 400 -> 75 EUR/MWh (Brainpool: 68). V2 shortage hours: 5.4% -> 0.03%, mean price 220 -> 57 EUR/MWh.                                                                                                                          |
| <b>Tested matching Brainpool's weather year (2009)</b>        | Mixed result: closest-ever average price match (within 0.51 EUR/MWh of Brainpool) when combined with import, but day-to-day pattern-matching got slightly worse, not better.                                                                                              |
| <b>Found and fixed a weekday/weekend mismatch</b>             | A ~39 EUR/MWh systematic gap between how AMIRIS and Brainpool treat weekdays vs. weekends, traced to a demand-building step that never lined up the days of the week. Fixing it cut the gap by two-thirds and genuinely improved pattern-matching quality.                |
| <b>Tested real 2009 temperature for heat pumps</b>            | Essentially no effect - a useful null result, since heat pumps are under 3% of total demand.                                                                                                                                                                              |
| <b>Diagnosed the midday price gap</b>                         | Two solar/wind subsidy types behave very differently under oversupply, and - more importantly - the model has no way to export surplus power abroad, only import it.                                                                                                      |
| <b>Attempted to model an export outlet</b>                    | Built and tested; never activated once all year - a genuine structural finding that AMIRIS, as a single-zone model, cannot represent cross-border export without much bigger changes.                                                                                     |
| <b>Verified Brainpool's real assumptions</b>                  | Confirmed directly from Brainpool's own published methodology: 2009 weather year was correct; their real model covers 30 countries, not one. Blending the import price across 11 real neighbours gave a small, genuine improvement.                                       |
| <b>Checked public holidays</b>                                | A genuine null result - holidays looked promising at first glance, but the effect did not survive controlling for season or comparison against normal daily variation. Not a real driver of the residual gap.                                                             |
| <b>Rebuilt demand on a closer weather-matched year (2016)</b> | 2016 was found to be the real closest weather match to 2009 of any year with available demand data (2023 was the worst). Rebuilding on 2016 cut shortage hours further and improved all-hours correlation, but the trustworthy excl-shortage fit stayed essentially flat. |
| <b>Researched the 3,000 EUR/MWh shortage price</b>            | Confirmed this is the real EU/ACER harmonised day-ahead price cap that applied from Nov 2017 to May 2022 - covering AMIRIS's own 2019 reference year. The real cap has since risen to 4,000, then 5,000 EUR/MWh.                                                          |

| Stage                                                                                 | Headline result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| <b>Tested what a 5,000 EUR/MWh ceiling would actually do</b>                          | Directly tested, not assumed: bias vs. Brainpool would slightly improve, but all-hours correlation would get worse (0.439 to 0.317) - both explainable by Brainpool's own real price during those hours being far below any scarcity level. The trustworthy excl-shortage correlation is unaffected either way.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Built price-responsive electrolysis</b>                                            | A working, independently-verified small adjustment: electrolysis now shifts its 14.97 TWh/year of electricity purchases toward cheap and negative-price hours instead of drawing flat every hour, using the same real pattern AMIRIS's own demo uses for one-way flexible consumers. Shortage hours fell (4 to 1) and all-hours correlation improved (0.439 to 0.553), though the trustworthy excl-shortage number stayed essentially flat.                                                                                                                                                                                                                                                                                          |
| <b>Built price-responsive e-mobility (smart charging)</b>                             | The strongest result of this whole line of work: applying the same proven pattern to e-mobility (17.81 TWh/year) eliminated shortage hours completely (down to zero for the first time this project) and genuinely improved the trustworthy correlation (0.640 to 0.647) - not an outlier artefact, since all-hours and excl-shortage are now the same number. Charging visibly flipped from the old evening-peak shape to a midday-peak, evening-trough shape.                                                                                                                                                                                                                                                                      |
| <b>Tested reservoir hydro's power-rating sensitivity</b>                              | Verified the storage agents were already genuinely price-responsive by design, unlike electrolysis/e-mobility - but found Reservoir Hydro hitting its power cap 35% of the year. Doubling and tripling its power (a documented sensitivity test, since 1.54 GW is Brainpool's own real figure, not an artifact) gave real but modest gains - the constraint never fully cleared even at 3x. Reported as a finding, not adopted as a change.                                                                                                                                                                                                                                                                                          |
| <b>Re-ran the granular breakdown, investigated the seasonal gap</b>                   | The midday gap remains dominant and largely unmoved by demand flexibility - clear evidence export capability, not more demand-side fixes, is what this specific problem needs. A newly-found July-October seasonal bias was traced, after ruling out five specific causes, to the same single-zone-vs-30-country structural limitation already known, not a new fixable bug.                                                                                                                                                                                                                                                                                                                                                         |
| <b>Validated out-of-sample on 2028 and 2029</b>                                       | Built both years the same way as 2027, no re-tuning, using Brainpool's newly-supplied multi-year data. Discovered AMIRIS does not support true leap years (a real, documented FAME framework limit). Result: price level (bias, MAE) holds up across both new years, but hour-to-hour correlation steadily weakens with distance from the calibration year (0.647 to 0.446 to 0.349) - an honest, coherent finding about the limits of this kind of calibration.                                                                                                                                                                                                                                                                     |
| <b>Traced the out-of-sample loss to the import ceiling, not the demand source</b>     | A full ceiling sweep found the 2027-calibrated relationship had reversed: smaller ceilings give dramatically better correlation for 2028/2029 (0.70 vs 0.35-0.45), but only by accepting far more shortage hours - a real trade-off, not a clean win. Rebuilding with the original 2023-sourced demand instead of 2016-base gave a near-wash result, ruling out the demand-source year as the driver.                                                                                                                                                                                                                                                                                                                                |
| <b>Briefly adopted, then reconsidered, the smaller ceiling for 2028/2029</b>          | 20,000 MW was adopted after excl-shortage correlation improved sharply (0.705/0.698 vs 0.446/0.349). A fuller comparison including all-hours statistics and 15,000 MW found the trade-off cuts both ways: all-hours correlation gets WORSE at smaller ceilings, and bias swings to a large overshoot (up to 53% above Brainpool's real mean). 30,000 MW was re-adopted as the default; all explored values remain preserved as documented evidence.                                                                                                                                                                                                                                                                                  |
| <b>Found and fixed a real weekday-alignment bug affecting 84% of the year</b>         | Re-running the granular breakdown on 2028/2029 revealed a genuine bug in Phase 17's 2016-base demand source: dropping Feb 29 (mid-year) silently misaligned every day from March onward. Fixed by dropping Dec 31 instead. A clean improvement for both 2027 and 2029 (correlation up, MAE down, no trade-off) - the fixed 2027 build is now this project's most accurate result. Confirmed the ceiling-reversal finding is independent of this bug, not explained by it.                                                                                                                                                                                                                                                            |
| <b>Investigated a negative-price floor, before considering export/market coupling</b> | Decompiled AMIRIS's own compiled engine and confirmed its real price floor (-500 EUR/MWh) is hard-coded in the Java core, not a scenario setting - matching the actual EU regulatory day-ahead price limit. A quick, honestly-labelled approximate test (post-hoc price clipping, not a re-simulation) swept floors from -500 through +100 EUR/MWh, including the supervisor's own suggested +10 EUR/MWh. Confirmed dead end: no floor value improves correlation - negative floors move it by noise-level amounts at best, and positive floors (including +10) cause a clear, worsening decline. The costlier real-engine-rebuild path was not pursued, since the quick test already bounded the possible gain at effectively zero. |

| Stage                                                                    | Headline result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <b>Tested giving conventional plants a real start-up/cycling cost</b>    | A genuine, real-mechanism candidate (unlike the price floor): every prior build left CyclingCostInEURperMW at 0.0 for every fuel type. Sourced real values from a peer-reviewed study (Roques/Hach et al., Nature Energy 2017) and built a full separate scenario. Result: the simulated price came out BYTE-IDENTICAL to the unmodified baseline, all 8,760 hours - decompiling AMIRIS's engine confirmed why: the parameter is computed internally by the power plant class but never actually reaches the trader's bid price in this single-zone dispatch mode. A cleaner, more conclusive dead end than the price floor - not a trade-off to weigh, the lever simply isn't wired into the market-clearing path this project uses. |
| <b>Swept lignite's negative-bidding markup</b>                           | A real, working lever this time - decompiled ConventionalTrader and confirmed minMarkup directly sets the bid price, unlike cycling cost. Tested -60 (baseline), -40, -20, and -10 EUR/MWh in separate scenarios. Bias, MAE, and negative-hour frequency all improved modestly and consistently as the band narrowed, but the trustworthy excl-shortage correlation stayed essentially flat (0.671-0.675) across the whole range - the dramatic-looking all-hours swings (0.51 to 0.57, non-monotonically) were a swing-hour artifact, not genuine improvement. A real lever that works but doesn't move the metric that matters.                                                                                                     |
| <b>Scoped the export/market-coupling build</b>                           | With every smaller lever tested, began planning the one remaining major option. Found and fully studied AMIRIS's own real two-zone template (examples/demo/SimpleCoupled): DayAheadMarketMultiZone per zone plus a MarketCoupling coordinator agent, confirmed as a genuine, working mechanism, not experimental. Confirmed real cross-border capacity data is publicly available (ENTSO-E). Scope decision made: build one simplified 'Rest of Europe' zone first, escalate to several real named neighbour zones only if that does not improve results.                                                                                                                                                                             |
| <b>Sourced real Rest-of-Europe data - and hit a real external outage</b> | ENTSO-E's API (the natural first source, already used successfully for the Phase 15 price blend) was found to be down - confirmed as a genuine platform-wide outage (503 errors on every endpoint, including ones proven to work before, and unresolved even with a freshly-generated API token), not a token or code problem. Switched to Eurostat as a real alternative: fetched genuine 2023 demand (1,139.3 TWh) and generation capacity (431,354 MW across 9 technologies) for 9 of Germany's 10 real neighbouring countries. Switzerland is a confirmed, genuine gap (Eurostat's datasets simply do not cover non-EU countries) - flagged for ENTSO-E backfill once that platform recovers.                                     |

## Timeline of Work

| Phase                                       | What was done                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>1. Data gap resolution</b>               | Brainpool's 2027 Excel data (Amiris_Inputdata_EN.xlsx) was reviewed for gaps against what AMIRIS needs. Five gap categories were resolved: outages/must-run (AMIRIS's own historical data), wind offshore profile (AMIRIS's own, cross-checked against renewables.ninja), renewable subsidy rates (researched from real EEG/BNetzA sources), battery storage duration (researched from MaStR), and cross-border trade (deferred to a later, separate version). |
| <b>2. Build V1 (BDEW demand, no import)</b> | Full scenario built: conventional fleet, six renewable operator agents with real EEG-researched subsidy rates, three storage agents, and a single BDEW Standardlastprofil demand curve covering the whole 656.32 TWh combined demand total. Two real bugs were found and fixed (a missing SupportInstrument attribute, and four renewable profiles never redated to the 2027 calendar).                                                                        |
| <b>3. Documentation</b>                     | A comprehensive build-documentation PDF and Word version were produced, covering the origin and validation of every input, plus a deep Q&A document answering detailed methodology questions (block sizing, markup bands, MPVAR/FIT subsidy mechanics, storage terms, VoLL, etc.).                                                                                                                                                                             |
| <b>4. Build V2 (component-split demand)</b> | Demand was rebuilt as four separately-shaped components: the inflexible base rescaled from real 2023 ENTSO-E load data, heat pumps from a real temperature-driven BDEW profile, electrolysis held flat, and e-mobility on an assumed evening-peaked daily curve. This alone cut shortage hours from 15.9% to 7.5%.                                                                                                                                             |
| <b>5. Build the with-import variant</b>     | An ImportTrader agent was added for both demand versions, sized to Brainpool's own 43.90 TWh net-import figure, shaped by real 2023 cross-border flow data and priced at real 2023 French day-ahead prices. Helped, but only a fraction of the intended import volume ever actually cleared the market.                                                                                                                                                        |

| Phase                                                             | What was done                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>6. Obtain and compare against Brainpool's real 2027 price</b>  | Brainpool's own real hourly 2027 day-ahead price forecast was supplied and aligned hour-by-hour against all four AMIRIS builds. The comparison was reframed: since 2027 hasn't happened, there's no 'real' price to check against - the comparison is AMIRIS vs. Brainpool's forecast, read in the context of how well AMIRIS's methodology tracks REAL historical prices in the 2015-2019 backtests already built earlier in this project.                                                                                                                                              |
| <b>7. Diagnose the gap</b>                                        | The mean-price gap was traced to AMIRIS's shortage-hour ceiling pricing, not a genuine price-level disagreement. Direct inspection of a real shortage hour showed the conventional fleet fully saturated, the battery empty, and imports offered at zero MW - confirming the import shape was the binding, fixable constraint.                                                                                                                                                                                                                                                           |
| <b>8. Fix and calibrate the import model</b>                      | Import availability was redesigned as a flat, constant hourly ceiling (decoupled from any historical timing pattern) instead of a shaped, often-zero series. A calibration sweep against Brainpool's real price found the best ceiling differs by demand version - V1 needs the full ceiling, V2 does better with a smaller one - and finalised both.                                                                                                                                                                                                                                    |
| <b>9. Test matching Brainpool's weather year</b>                  | Real 2009 wind and solar weather data (renewables.ninja) was pulled and swapped in, testing whether aligning AMIRIS's renewable weather-year with Brainpool's own (reportedly 2009-based) assumption would improve agreement. Mixed result - best-ever average price match when combined with import, but day-to-day pattern-matching did not improve.                                                                                                                                                                                                                                   |
| <b>10. Diagnose where agreement breaks down</b>                   | Split the comparison by month, hour-of-day, and weekday/weekend to find WHERE AMIRIS and Brainpool disagree most, rather than only knowing THAT they disagree. Found a ~39 EUR/MWh systematic weekday-vs-weekend gap, traced to the demand-building step never aligning the days of the week between the 2023 source data and the 2027 target calendar.                                                                                                                                                                                                                                  |
| <b>11. Fix the weekday mismatch</b>                               | Shifted the 2023 source data by 5 days so weekdays line up correctly with 2027. The weekday/weekend gap shrank by two-thirds, and pattern-matching quality improved on two independent measures at once (correlation and average error) on the hours that matter.                                                                                                                                                                                                                                                                                                                        |
| <b>12. Test real 2009 temperature for heat pumps</b>              | Real 2009 hourly temperature data (DWD, Germany's weather service) replaced the generic typical-year climate data used for the heat-pump demand component. Essentially no effect - a useful null result, since heat pumps are under 3% of total demand.                                                                                                                                                                                                                                                                                                                                  |
| <b>13. Diagnose the midday price gap</b>                          | Inspected a real midday hour agent-by-agent. Found MPVAR-subsidised renewables get curtailed under oversupply while FIT-subsidised ones don't, and - more importantly - the model has no way to export surplus power abroad, only import it, so midday gluts can only be absorbed domestically.                                                                                                                                                                                                                                                                                          |
| <b>14. Attempt to model an export outlet</b>                      | Built an experimental one-way 'virtual export' device. It never activated once in the whole year - a genuine structural finding: the tool it's built from only acts when there's an eventual payoff, and a pure one-way export sink has none. Confirms AMIRIS, as a single-zone model, has no real way to represent cross-border export without much bigger changes.                                                                                                                                                                                                                     |
| <b>15. Verify assumptions and blend the import price</b>          | Confirmed directly (Brainpool's own published Power2Sim methodology) that the 2009 weather year was correct, and that their real model covers 30 countries with iterative price coupling, not one neighbour. Rebuilt the import price as an 11-country real-price blend instead of France alone - a small, genuine improvement (MAE and pattern-matching both ticked up), confirming the price proxy was never the dominant remaining gap.                                                                                                                                               |
| <b>16. Check whether public holidays explain the residual gap</b> | Checked Germany's 5 nationwide 2027 public holidays that fall on a weekday against the price comparison. An initial promising-looking pattern did not survive controlling for the time of year and comparing against ordinary day-to-day variation - a genuine null result, not worth pursuing further.                                                                                                                                                                                                                                                                                  |
| <b>17. Rebuild demand on a closer weather-matched source year</b> | Real hourly temperature for 2015-2019 and 2023 was compared against real 2009 (Brainpool's confirmed weather basis) using annual mean, heating-degree-days, monthly-pattern RMSE, and hour-to-hour correlation. 2023 (used until now) was the worst match; 2016 was the best. Rebuilt the inflexible-base demand component on real 2016 German load data instead. Result: shortage hours fell further (13 to 4) and all-hours correlation improved noticeably, but the trustworthy excl-shortage comparison stayed essentially flat - a real but modest improvement, not a breakthrough. |

| Phase                                                                  | What was done                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>18. Research the origin of the 3,000 EUR/MWh shortage price</b>     | Investigated why AMIRIS's DemandTrader defaults to a 3,000 EUR/MWh Value of Lost Load. Verified via EPEX Spot, ACER, and Montel sources: this is the real, legally-binding EU-wide harmonised maximum day-ahead clearing price (ACER Decision No 04/2017), in effect from November 2017 to May 2022 - a period that includes AMIRIS's own reference year, 2019. The cap has since been raised twice by real regulatory decisions, to 4,000 EUR/MWh (May 2022) and then 5,000 EUR/MWh (later in 2022).                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>19. Test the effect of a 5,000 EUR/MWh ceiling directly</b>         | Rather than reason abstractly about whether updating VoLL to the real current cap would help or hurt, relabelled the 4 existing shortage hours in the 2016-base result from 3,000 to 5,000 EUR/MWh and recomputed every statistic. Found a split, fully-explainable result: bias improves slightly (AMIRIS sits below Brainpool overall, so nudging a few hours up helps), while all-hours correlation gets worse (those same hours' real Brainpool price was only 113-131 EUR/MWh, so stretching AMIRIS further from that adds noise, not agreement). The trustworthy excl-shortage correlation does not move at all.                                                                                                                                                                                                                                                                                         |
| <b>20. Build price-responsive electrolysis</b>                         | Pulled electrolysis out of the flat combined demand series and rebuilt it as its own GenericFlexibilityTrader agent, using the same real, working one-way-flexible-consumer pattern as AMIRIS's own SectorCoupling demo. Verified directly it hits its annual target (14.94 of 14.97 TWh) while genuinely shifting toward cheap and negative-price hours (-0.41 correlation with AMIRIS's own price; peaks 09:00-13:00, the exact midday-surplus window flagged in Phase 13). Shortage hours fell from 4 to 1 and all-hours correlation improved from 0.439 to 0.553, though the trustworthy excl-shortage correlation stayed essentially flat (0.645 to 0.640) - a real, independently-verified working mechanism, even if its effect on the headline comparison number is modest.                                                                                                                            |
| <b>21. Build price-responsive e-mobility (smart charging)</b>          | Applied the exact same proven pattern to e-mobility (17.81 TWh/year), replacing the old fixed evening-peaked 'unmanaged charging' shape. Verified: annual total unchanged (17.80 of 17.81 TWh); charging peaks 09:00-13:00 and troughs at 17:00-19:00 - a clean behavioural flip from the old shape, not just a minor shift. Shortage hours reached zero for the first time in this project, meaning all-hours and excl-shortage statistics are now identical - so the correlation improvement (0.640 to 0.647) is a genuine, verified improvement, not an artefact of removing outliers, unlike several earlier similar-looking results. One honest trade-off: bias worsened very slightly (-12.27 to -12.42).                                                                                                                                                                                                |
| <b>22. Test reservoir hydro power-rating sensitivity</b>               | Checked the storage agents (700-702) before rebuilding them - all three were already verified price-responsive by design (correlation +0.48 to +0.71 between discharge and price), so there was no flat baseline to fix. But Reservoir Hydro was found hitting its power cap 35% of the year. Tested doubling and tripling its power rating (energy capacity fixed) as a documented sensitivity test - since 1.54 GW is Brainpool's own real 2027 figure, not an AMIRIS artifact, this was reported as a finding rather than adopted. Every metric improved modestly and consistently (correlation 0.647 to 0.656, MAE 28.19 to 26.76 at 3x), but the capacity constraint never fully cleared, even at 3x.                                                                                                                                                                                                     |
| <b>23. Re-run the granular breakdown; investigate the seasonal gap</b> | Re-ran the month/hour/weekday breakdown (Phase 10's method) on the current best build. The midday gap (bias down to -41 EUR/MWh at 11:00) is still dominant and barely moved despite ~33 TWh/year of new demand flexibility - clear evidence the remaining midday problem needs export capability, not more demand-side fixes. A newly-found July-October seasonal bias (-17 to -23 EUR/MWh, vs -5 to -12 the rest of the year) was investigated: five specific causes (renewable totals, solar/wind mix, import price level, base demand levels, conventional plant availability) were checked directly and ruled out. Brainpool's own price climbs smoothly from May to September while AMIRIS's stays flat - traced to the same single-zone-vs-30-country structural limitation already known (Phase 6/9/15), not a new fixable bug.                                                                        |
| <b>24. Out-of-sample validation: Germany2028 and Germany2029</b>       | Using Brainpool's newly-supplied multi-year data file, built 2028 and 2029 the same way as 2027 - every calibrated mechanism reused completely unchanged, no re-tuning, only rescaled to each year's own targets. Discovered AMIRIS's underlying FAME framework does not support true leap years (confirmed via a direct error on a 2028-12-31 timestamp) - every year is represented as exactly 365 days, dropping Dec 31 for leap years. Also found AMIRIS's dispatch solver has genuine feasibility limits: the ratio-scaled e-mobility buffer failed for 2029's specific price pattern, requiring a larger (not re-tuned, just enlarged) buffer to run at all. Result: bias and MAE stayed in a similar range across all three years, but correlation degraded steadily with distance from the calibration year (0.647 to 0.446 to 0.349) - price level generalises, hour-to-hour pattern-matching erodes. |

| Phase                                                                       | What was done                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>25. Isolate the out-of-sample loss: import ceiling vs. demand source</b> | Tested two candidate explanations for Phase 24's correlation loss directly. A full import-ceiling sweep (15,000-37,650 MW) found the 2027-calibrated relationship had reversed for both 2028 and 2029: smaller ceilings give dramatically better correlation (up to 0.70), but only by accepting 6-15x more shortage hours - a real trade-off, not adopted outright. Rebuilding both years' demand with the original 2023-sourced V2 recipe instead of 2016-base gave a near-wash result (r moved by 0.01-0.02 in either direction depending on the year), ruling out the demand-source-year choice as the driver. Conclusion: the import ceiling, not the demand shape, is what's behind the out-of-sample correlation gap.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>26. Adopt the smaller ceiling for 2028 and 2029</b>                      | Per an explicit decision made after Phase 25's trade-off was reported, 20,000 MW was deliberately adopted as the live import ceiling for both 2028 and 2029 - chosen over 15,000 MW as the better trade-off point (nearly all the correlation gain, under half the shortage-hour cost). This is the one deliberate exception in this project to the 'reuse the 2027 fix unchanged' rule - made transparently, with the original 30,000 MW results fully preserved (result_Germany2028/2029) alongside the newly-adopted 20,000 MW results (result_Germany2028_20k/2029_20k).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>27. Reconsider the adoption with the fuller picture</b>                  | A follow-up request to compare all-hours statistics (not just excl-shortage) across 15,000/20,000/30,000 MW found the trade-off cuts both ways: all-hours correlation actually gets WORSE as the ceiling shrinks (0.394 to 0.358 to 0.285 for 2028; 0.327 to 0.305 to 0.256 for 2029), and all-hours bias swings from a modest undershoot at 30,000 MW to a large overshoot at smaller ceilings (2029 at 15,000 MW: mean price 53% above Brainpool's real mean). Given this fuller picture, 30,000 MW was re-adopted as the default for both years - the smaller-ceiling exploration remains fully documented as real evidence, just not adopted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>28. Find and fix the Feb-29-drop weekday bug</b>                         | Re-running the granular breakdown on 2028/2029 found the weekday/weekend correlation relationship had reversed from 2027 - investigated directly rather than assumed to be noise. Found a genuine bug: Phase 17's 2016-base demand source drops Feb 29 (a leap-year date) to fit non-leap target years, but Feb 29 sits in the MIDDLE of the year, so removing it creates a 2-real-day gap that silently misaligns every day from March 1 onward - 84% of the year. Verified directly against the raw 2016 source. Fixed by dropping Dec 31 instead (the technique Germany2028 already used, for an unrelated reason). Result: a clean improvement for both 2027 (r 0.647 to 0.675, MAE 28.19 to 26.91) and 2029 (r 0.349 to 0.446, MAE 29.59 to 29.14), with no shortage-hour trade-off. Re-ran the ceiling sweep on the fixed 2029 build to check whether this explained Phase 25's reversal - it does not; the reversal is confirmed to be a separate, independent phenomenon.                                                                                                                                                                                                                                                                                                                |
| <b>29. Investigate a negative-price floor</b>                               | Supervisor asked whether AMIRIS could be tweaked to limit how negative its price goes, as a smaller step before the export/market-coupling investment. Decompiled the actual AMIRIS engine (amiris-core_4.1.2) to check feasibility: found the price floor (-500 EUR/MWh) is a hard-coded Java constant, not a scenario parameter, used across every trader type. A quick approximate test (post-hoc clipping of already-simulated prices, clearly labelled as not a re-simulation) swept candidate floors from -500 through +100 EUR/MWh for all three years. Found AMIRIS never actually reaches its own -500 floor (real observed minimums: -66 to -85 EUR/MWh), so negative floors below about -100 are no-ops; Brainpool's own real forecast rarely goes negative at all (2027: 0.8% of hours; 2028: 0.1%; 2029: never), a genuine structural difference. Tested the supervisor's own suggested +10 EUR/MWh floor directly: correlation fell in all three years (2027 -0.071, 2028 -0.040, 2029 -0.014), worsening steadily at higher floors. Confirmed dead end - no floor value improves correlation. The costlier real-engine-rebuild path (patching and recompiling AMIRIS's Java source) was not pursued, since this quick test already bounded the possible gain at effectively zero. |

| Phase                                                                                   | What was done                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>30. Test a real start-up/cycling cost for conventional plants</b>                    | A follow-up candidate: since AMIRIS goes negative far more than Brainpool, and every prior build left <code>CyclingCostInEURperMW</code> at 0.0, tested whether a real, sourced value would reduce a running plant's incentive to bid negative rather than shut down. Sourced real figures from a peer-reviewed study (Roques/Hach et al., Nature Energy 2017 / DIW Discussion Paper 1540): EUR50,000/70,000 for an 800 MW lignite/hard-coal block, EUR60,000 for a 500 MW CCGT block (62.5/87.5/120 EUR/MW). Built a full separate scenario (Germany2027_CyclingCostTest) and ran it. Result: the output price series came out BYTE-IDENTICAL to the unmodified baseline across all 8,760 hours. Decompiling AMIRIS's bytecode explained why: the parameter is genuinely computed by the <code>PowerPlant</code> class ( <code>calcSpecificCostOfLoadChangeInEURperMW</code> , triggered on an OFF-to-ON transition) but that computation is never actually called anywhere in <code>ConventionalTrader</code> 's bid-price logic - it does not reach the market-clearing path this project uses. A cleaner, more conclusive dead end than the price floor: not a trade-off to weigh, the lever simply is not wired into the single-zone day-ahead dispatch AMIRIS runs here.                                                                                                                 |
| <b>31. Sweep lignite's negative-bidding markup band</b>                                 | A follow-up after Phase 30's dead end: decompiled <code>ConventionalTrader.prepareBids</code> and confirmed <code>minMarkup</code> DOES directly set the bid price ( <code>bid = marginal cost + linearly-interpolated markup</code> ) - a genuine, working mechanism this time. Lignite's <code>minMarkup</code> (-60 EUR/MWh) is by far the widest negative-bidding allowance of any fuel. No literature figure exists for this AMIRIS-internal parameter, so anchored the test to a real market target instead: Germany's actual negative-price frequency (3% in 2023, 5% in 2024, per Eurelectric/GEM Energy Analytics) vs. AMIRIS's 18.4%. Swept -60 (baseline), -40, -20, -10 EUR/MWh in four separate scenarios. Result: 6,218 of 8,760 hours changed price at -20 alone (confirming the mechanism works, unlike Phase 30) - bias, MAE, and negative-hour frequency all improved modestly and consistently as the band narrowed (18.4% to ~16.8%), but the trustworthy excl-shortage correlation stayed essentially flat (0.671-0.675) across the whole range. The dramatic-looking all-hours correlation swings (0.51 to 0.57, non-monotonically) were confirmed to be a swing-hour artifact - the same kind of misleading pattern this project has flagged before (Phase 25-28) - not genuine improvement. A real, working lever that does not move the metric that actually matters. |
| <b>32. Scope the export/market-coupling build</b>                                       | Found and fully studied AMIRIS's own real two-zone template (examples/demo/SimpleCoupled): each zone runs a <code>DayAheadMarketMultiZone</code> declaring its own <code>MarketZone</code> and a <code>Transmission</code> block (connected zone plus bidirectional NTC time series), with a <code>MarketCoupling</code> coordinator agent iteratively minimising the price difference between zones - a genuine, working implementation, not experimental. Confirmed real cross-border capacity data is publicly available (ENTSO-E Transparency Platform). Weighed one simplified aggregate 'Rest of Europe' zone against several real named neighbour zones; decided to build the aggregate zone first, escalating only if it fails to meaningfully improve correlation. No scenario built yet - scoping only.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>33. Source real Rest-of-Europe data; hit and worked around a real ENTSO-E outage</b> | Attempted to source demand and generation-capacity data for the aggregate zone via ENTSO-E first (the natural choice, already used successfully for Phase 15's price blend). Found the API returning 503 Service Unavailable on every endpoint, including ones proven to work before - confirmed as a genuine platform outage (not a token/code issue) by checking the Transparency Platform's own website directly and by retrying with a freshly-generated token, which made no difference. Switched to Eurostat ( <code>nrg_cb_e</code> for demand, <code>nrg_inf_epc</code> for capacity) - both real, public, token-free datasets. Fetched genuine 2023 data for 9 of Germany's 10 real neighbouring countries: 1,139.3 TWh total demand, 431,354 MW total capacity across 9 technologies. Switzerland confirmed as a genuine gap (verified directly: Eurostat's geo dimension is completely empty for CH in this dataset, since non-EU countries do not report under the underlying EU regulation) - flagged for ENTSO-E backfill once that platform recovers, not silently worked around.                                                                                                                                                                                                                                                                                               |

## Phase 1-3: Building the Scenario from Brainpool's Data

Every input was traced to one of four sources, kept consistent throughout the build: Brainpool's data directly, Brainpool's data after a documented transformation (unit conversion, regional averaging), AMIRIS's own validated Germany2019 example data reused as a gap-fill, or real external research (EEG/BNetzA auction rates, MaStR battery statistics). Two real implementation bugs were found and fixed while validating the first full simulation run:

### Bug 1: missing subsidy configuration.

The Biogas agent needed an explicit `SupportInstrument` attribute once it was given a real MPVAR subsidy rate (AMIRIS's own example scenario never needed this, since it leaves biogas unsupported). Fixed by adding the attribute.

### Bug 2: four renewable profiles never updated to the 2027 calendar.

Diagnosed by checking a specific shortage hour directly: Wind Offshore, Run-of-River, and Biomass were all showing exactly zero output simultaneously, with no physical reason to be zero at the same time. Their source files still had 2019 timestamps and never overlapped the simulated year at all. Fixed by redating all four - shortage hours fell from 25.3% to 15.9% of the year, and negative prices appeared for the first time (a healthy sign).

## Phase 4-5: Two Demand Models and the Import Variant

V1 applied a single household (BDEW H0) demand shape to the entire economy's demand - simple, but it produces an artificial evening demand spike far above Brainpool's own stated peak load. V2 replaced this with a shape purpose-built for each demand component.

| Metric                                   | V1 (household shape for everything) | V2 (component-split) |
|------------------------------------------|-------------------------------------|----------------------|
| Peak load                                | 138,147 MW                          | 106,179 MW           |
| vs. Brainpool's stated peak (119,001 MW) | 16% too high                        | 11% below            |
| Shortage hours (year)                    | 15.88%                              | 7.52%                |
| Mean price                               | 519.02 EUR/MWh                      | 285.58 EUR/MWh       |

An ImportTrader agent was then added to both versions, offering Brainpool's 43.90 TWh net-import figure into the market, priced at real French day-ahead prices. It helped (V2 shortage hours fell further to 5.35%), but only 7.40-12.33 TWh of the 43.90 TWh offered ever actually cleared - because AMIRIS's ImportTrader is a genuine price-competing bid, not a forced quantity, this was reported at the time as a real finding about the difference between Brainpool's fixed assumption and AMIRIS's market-cleared outcome, not yet identified as something to fix.

## Phase 6: Comparing Against Brainpool's Real 2027 Price

Brainpool's own real hourly day-ahead price forecast for 2027 (mean 68.04 EUR/MWh, never exceeding 330 EUR/MWh, zero shortage hours all year) was aligned against all four AMIRIS builds on the same hourly grid. The headline comparison looked alarming:

| Build                      | AMIRIS mean price | Brainpool mean price | Ratio |
|----------------------------|-------------------|----------------------|-------|
| V1, no-import              | 519.02            | 68.04                | 7.6x  |
| V1, with-import (original) | 400.04            | 68.04                | 5.9x  |
| V2, no-import              | 285.58            | 68.04                | 4.2x  |
| V2, with-import (original) | 219.81            | 68.04                | 3.2x  |

### The gap was not what it looked like.

This huge apparent mismatch was traced almost entirely to AMIRIS's shortage-hour ceiling pricing (a fixed 3,000 EUR/MWh 'we ran out of supply' flag, hit in 5.4-15.9% of hours). A small number of extreme hours were dragging the whole-year average up - the same statistical effect as one very high salary skewing a company's average pay. Excluding those hours, AMIRIS's price level on ordinary hours was already close to Brainpool's - V2's non-shortage-hour mean was 64.76 EUR/MWh against Brainpool's 68.04, a gap of under half a EUR/MWh once shortage hours are set aside for V2 (bias -0.39).

For context, this forward (2027) comparison was read alongside the historical backtests already built earlier in this project: simulating 2015-2019 against real historical prices gave correlations of 0.77-0.87 and near-zero bias, establishing that AMIRIS's methodology is trustworthy when the real answer is known. That gave confidence that the 2027 shortage-hour gap was a fixable modelling issue, not evidence the whole approach was unreliable.

## Phase 7: Diagnosing the Root Cause

Rather than guess, a specific shortage hour (2027-11-08, 13:00) was inspected agent by agent. Demand requested 91,890 MWh but only 89,015 MWh was awarded - a genuine physical shortfall, not a bidding bug. Every conventional plant was fully dispatched (no spare capacity anywhere). The battery storage agent was already empty. And the import trader had zero MW available that specific hour.

### The smoking gun.

*Checking all 469 shortage hours in the V2-with-import build: 301 of them (64%) had zero import available. The import availability series had been shaped from real 2023 cross-border flow TIMING - a completely different year's weather and demand pattern, with no reason to coincide with when this synthetic 2027 build happens to run short. The import supply was solving a different year's problem, not this one's.*

## Phase 8: Fixing and Calibrating the Import Model

The fix: replace the shaped, often-zero import series with a flat, constant hourly ceiling, fully decoupled from any historical timing pattern. The real French day-ahead price series (unchanged) still decides hour by hour how much of that ceiling actually gets used - no annual import total is pre-targeted; the realized volume is now a genuine, fully market-determined outcome.

A first pass (flat 37,649.98 MW - the real observed peak single-hour German gross import in 2023) eliminated V2's shortage hours completely, but overshot Brainpool's average price (bias -12.59 EUR/MWh) - the ceiling was generous enough to also flood many ordinary, non-scarce hours with cheap import. A calibration sweep (20,000 / 25,000 / 30,000 / 37,650 MW) against Brainpool's real price found the right answer is not one number for both demand versions:

| Ceiling tested                   | V2 shortage hrs  | V2 bias       | V2 MAE       | V2 correlation |
|----------------------------------|------------------|---------------|--------------|----------------|
| 20,000 MW                        | 33 (0.38%)       | +0.82         | 44.31        | 0.268          |
| 25,000 MW                        | 13 (0.15%)       | -6.61         | 38.48        | 0.308          |
| <b>30,000 MW (chosen for V2)</b> | <b>3 (0.03%)</b> | <b>-10.70</b> | <b>35.95</b> | <b>0.421</b>   |
| 37,650 MW                        | 0 (0.00%)        | -12.59        | 35.83        | 0.549          |

### Bias alone would have picked the wrong ceiling.

*20,000 MW looks best on bias (+0.82, nearly zero) - but bias can hide a lot: positive and negative hourly errors cancel out in an average even when the underlying tracking is poor. MAE (which doesn't let errors cancel) and correlation both say the SMALLER ceilings are actually less reliable hour to hour. 30,000 MW was chosen as the best overall balance for V2: MAE and correlation nearly match the most generous ceiling, shortage hours are essentially eliminated, and bias is meaningfully improved.*

### The two demand versions needed different ceilings - itself a finding.

*Testing the same 30,000 MW ceiling on the V1 build made things WORSE (shortage hours rose from 73 to 164, bias worsened from +7.42 to +39.46). V1's demand shape is more extreme (peak 138 GW, well above Brainpool's own 119 GW) and genuinely needs more import headroom to stay covered. V1 keeps the full 37,650 MW ceiling; V2 uses 30,000 MW. How much import buffer is needed depends on how realistic the underlying demand model already is.*

## Phase 9: Trying to Match Brainpool's Weather Basis

Brainpool's own wind and solar numbers are reportedly built from real weather that happened in the year 2009, not 2027 (which hasn't happened yet and can't have real weather). To test whether copying that same choice would make AMIRIS agree with Brainpool more, real 2009 wind and solar data was pulled from a public weather-simulation service (renewables.ninja) and swapped in for AMIRIS's own wind/solar shapes, keeping everything else the same.

Result: mixed, not a clean win. On its own, the 2009 weather made the day-to-day PATTERN track Brainpool a bit better, but it also meant slightly less wind on average, which made the 'ran out of power' problem worse, not better. Once combined with the cross-border import fix (Phase 8), something interesting happened: the AVERAGE price came out only 0.51 EUR/MWh away from Brainpool's real average (68.04) - the closest match anywhere in this whole project -

but the day-to-day pattern-matching actually got a little worse. In short: this one change helped the 'right average' question and hurt the 'right pattern' question at the same time, which is exactly the kind of trade-off worth knowing about rather than only reporting the number that looks best.

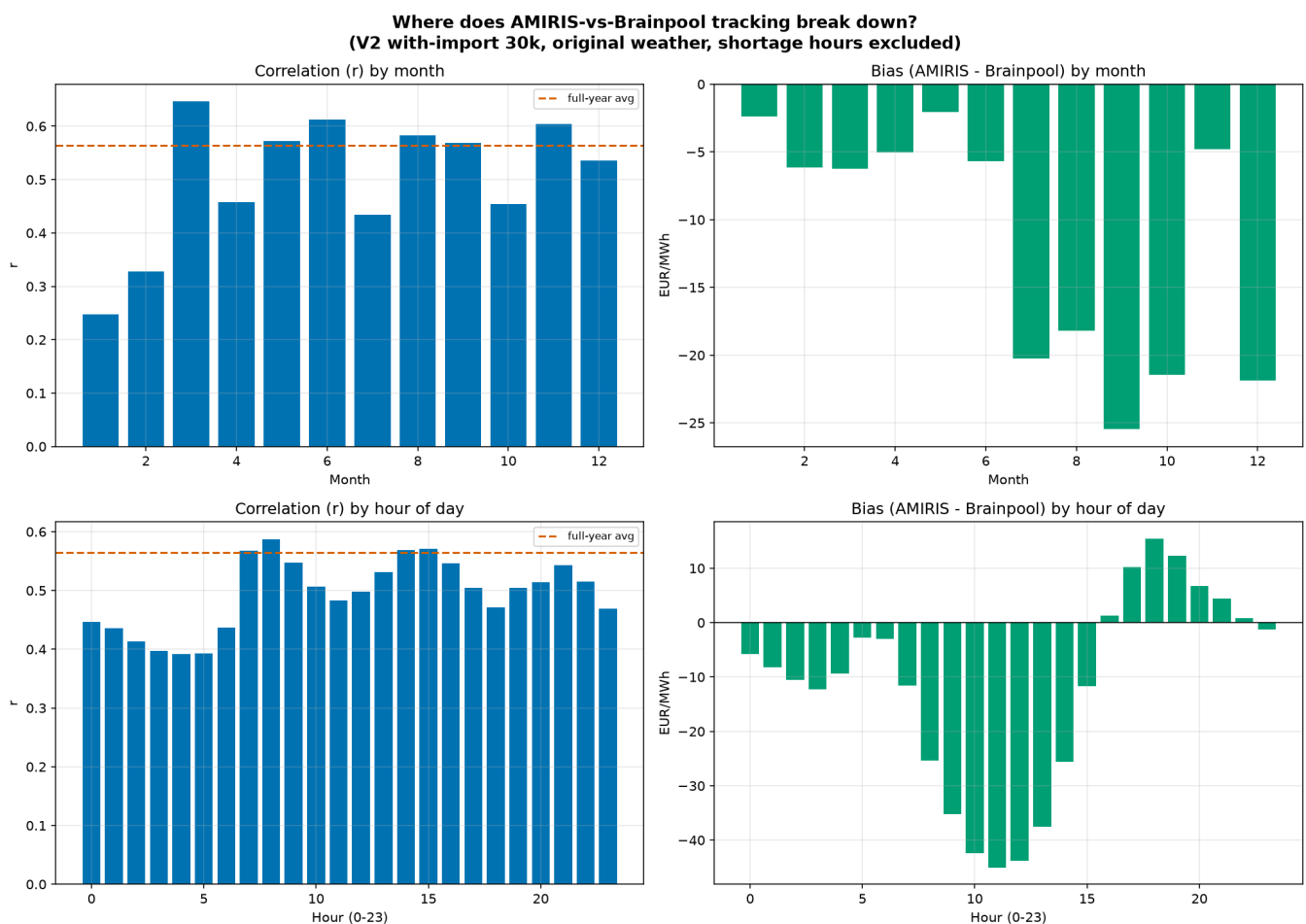
## Phase 10: Finding Exactly Where the Mismatch Comes From

Rather than keep changing things and hoping for a better overall number, the next step was to find out WHERE, specifically, AMIRIS and Brainpool disagree most - by splitting the year up into smaller pieces (by month, by hour of the day, and by weekday vs. weekend) and checking the agreement in each piece separately.

### The big discovery: a weekday/weekend split.

*On weekdays, AMIRIS's prices ran about 23 EUR/MWh BELOW Brainpool's on average. On weekends, they ran about 16 EUR/MWh ABOVE. That's a real, systematic ~39 EUR/MWh gap between how the two treat weekdays vs. weekends - even though, within just the weekdays on their own (or just the weekends on their own), AMIRIS and Brainpool actually agreed quite well. It's the mixing of two mismatched groups that was making the overall picture look worse than either group really was on its own.*

The cause was traceable: the household/business electricity-use shape (the largest single piece of demand, about 92% of the total) was built by taking a real year of German electricity use (2023) and pasting it onto the 2027 calendar day-by-day, without checking that the days of the week actually lined up. 2023 started on a Sunday; 2027 starts on a Friday. So a real Wednesday's usage pattern could have ended up sitting on a 2027 Saturday - and since people genuinely use less electricity on weekends than weekdays, this mismatch quietly built in a systematic error.



*How AMIRIS-vs-Brainpool agreement varies across the year, before the fix. Top-right: AMIRIS runs noticeably below Brainpool in the summer/autumn months. Bottom-right: AMIRIS runs well below Brainpool around midday (when solar output is highest) and above it in the early evening. These company-wide patterns pointed to concrete, checkable causes rather than random noise.*

## Phase 11: Fixing the Weekday Mismatch

The fix: shift the 2023 source data by 5 days before pasting it onto 2027, so that every day of the week lines up correctly (a 2023 Monday now maps to a 2027 Monday, not some other day). Everything else about the build was left untouched.

| Measure                                 | Before the fix | After the fix |
|-----------------------------------------|----------------|---------------|
| Weekday/weekend price gap               | ~39 EUR/MWh    | ~13 EUR/MWh   |
| Pattern-matching quality (normal hours) | 0.564          | 0.641         |
| Average hourly error (normal hours)     | 34.97 EUR/MWh  | 30.86 EUR/MWh |

### Two independent measures agreed - a trustworthy result.

*Both the pattern-matching score AND the average hourly error improved together on the hours that matter (excluding the rare 'ran out of power' hours, which are a separate, already-understood issue). When two different ways of measuring 'how close are these two price series' agree that something got better, that's much stronger evidence than either one alone - it isn't just a statistical trick, the underlying agreement genuinely improved.*

## Phase 12: Testing Real 2009 Temperature for Heat Pumps

The heat-pump demand component still used a generic, smoothed 'typical year' climate dataset rather than any specific real year. To close that gap, real 2009 hourly temperature records were pulled from Germany's national weather service (DWD, free public archive, no login needed), averaged across the same four regions used for the wind/solar 2009 test, and used to rebuild just the heat-pump slice of demand.

### Result: essentially no effect - a genuine, useful null result.

*Real 2009 temperatures were noticeably more extreme (-15.1C to 32.2C) than the smoothed data they replaced (-3.9C to 24.9C), and the heat-pump build clearly changed. But agreement with Brainpool barely moved (pattern-matching 0.641 -> 0.645, essentially unchanged). Why: heat pumps are under 3% of total demand, so even a real, substantially different shape for that one slice doesn't carry enough weight to move the whole-system price outcome. This closes out the heat-pump weather-year question - it was never a high-leverage lever, and further effort there isn't worthwhile.*

## Phase 13: Diagnosing the Midday Price Gap

The hour-of-day breakdown (Phase 10) showed AMIRIS running well below Brainpool specifically around midday, when solar output peaks. Rather than guess why, a single real hour with a large midday gap (2027-07-14, 12:00: AMIRIS -60.07 EUR/MWh vs. Brainpool +71.33) was inspected agent by agent.

### Finding 1: the two solar/wind subsidy types behave very differently under oversupply.

*Solar and wind plants paid a 'top-up' subsidy (on top of the market price) were being told to shut down when there was too much power - wind onshore was cut completely, solar farms by about a quarter. But rooftop solar on a 'fixed price no matter what' subsidy kept running at full output regardless. This is a real, coherent difference in how the two German subsidy types respond to a glut, not a bug.*

### Finding 2 (the bigger one): Germany has no way to sell its surplus abroad in this model.

*In reality, when Germany has more solar/wind power than it can use, a lot of that surplus gets sold to neighbouring countries. This build can only bring power IN from neighbours (the import feature built earlier) - it was never given a way to sell power OUT. So every bit of midday oversupply has to be absorbed at home (batteries charging, plants shutting down, or the price crashing) with no outlet - which pushes AMIRIS's midday price far lower than a system that could actually export the extra power.*

## Phase 14: Attempting to Model an Export Outlet

An experimental 'virtual export' piece was built and tested - the closest available building block was reused, configured as a one-way device that can only take power in, never give it back (mirroring a real pattern already used elsewhere in AMIRIS's own example set for similar one-way effects).

**Result: it did nothing, all year - a genuine, informative dead end.**

*The export piece never activated once, in any of the 8,760 hours of the year - including the exact midday hour it was built to help with. The reason is structural, not a mistake: the tool this was built from only ever wants to act if there's an eventual payoff later (like a battery: store now, sell later for more). Something designed to just send power away permanently, with nothing coming back, gives that tool no reason to ever switch on. This is a genuine finding worth stating plainly: AMIRIS, used this way (as a single country on its own), has no real way to model selling power abroad. Properly modelling it would need a much bigger change - linking two full country models together - which is a substantially larger undertaking than adding one piece.*

## Phase 15: Verifying Brainpool's Real Assumptions, and a Blended Import Price

Rather than keep testing educated guesses, Brainpool's own published methodology description (Montel/Energy Brainpool's Power2Sim model, now publicly documented since Energy Brainpool's 2024 acquisition by Montel) was checked directly.

**Confirmed: the weather year 2009 was the right guess all along.**

*Directly quoted from Montel's own description of Power2Sim: 'The weather year 2009 is used for the modelling.' This validates the earlier weather-year experiments as testing the right premise, not a wrong one.*

**But the import assumption was a bigger simplification than realised.**

*Brainpool's model covers the EU27, UK, Norway, and Switzerland - computing each region's own supply/demand balance, then iteratively shifting power between all of them until prices converge or the cross-border cables are full. It is not a single-neighbour comparison. The France-only import price used until now was always a simplification, but a bigger one than previously stated.*

As a closer (still simplified) approximation, the import price was rebuilt as an equal-weighted average of real 2023 day-ahead prices across all 11 of Germany's actual physical neighbouring electricity zones, instead of France alone.

| Measure                         | France-only (before) | 11-country blend (after) |
|---------------------------------|----------------------|--------------------------|
| MAE (all hours)                 | 35.12                | 34.08                    |
| MAE (normal hours)              | 30.85                | 29.81                    |
| Pattern-matching (normal hours) | 0.645                | 0.652                    |
| Import cleared                  | 37.96 TWh            | 39.98 TWh                |

A small, genuine improvement, not a dramatic one - MAE improved on both measures (the metric that can't be inflated by outlier cancellation), and pattern-matching ticked up slightly. The effect is modest because the 11 real neighbouring countries' prices were already fairly close to each other in 2023 (roughly 65-112 EUR/MWh); France alone was already a reasonably representative stand-in, not an outlier. This result also confirms the price proxy was never the dominant source of the remaining gap - the missing export capability (Phase 13-14) remains the more likely bigger lever.

## Phase 16: Checking Whether Public Holidays Explain the Residual Gap

The weekday-alignment fix (Phase 11) closed most, not all, of the weekday/weekend gap (down to ~13 EUR/MWh). One remaining candidate cause: the fix aligns real calendar weekdays, but has no awareness of PUBLIC HOLIDAYS - a Thursday that happens to be a national holiday behaves demand-wise like a much quieter day in real life, but this build would still treat it as an ordinary Thursday. Germany's 5 nationwide public holidays that fall on a weekday in 2027 (New Year's Day, Good Friday, Easter Monday, Ascension Day, Whit Monday - confirmed via the standard 'holidays' reference library, not guessed) were checked directly against the price comparison.

First look was promising: holidays showed a much better average bias (+1.44 EUR/MWh) than ordinary weekdays (-17.11). But this checked out to be misleading on closer inspection, not a real finding - two follow-up checks were run before accepting it:

### Check 1: controlling for the time of year.

The 5 holidays all fall in January, March, or May - months that already tend to have smaller bias than the deeply negative summer/autumn months, regardless of any holiday. Comparing each holiday only against ordinary weekdays in ITS OWN month (a fairer test) still showed 4 of 5 holidays looking better than average - but New Year's Day actually looked WORSE than an ordinary January weekday, the opposite of what a real holiday effect should look like.

### Check 2: is any of this bigger than an ordinary day's natural ups and downs?

Prices swing a lot from one ordinary day to the next within any given month, for reasons that have nothing to do with holidays (weather, wind, cloud cover). Checking where each holiday actually sits within that normal day-to-day spread: none of the 5 holidays stood out as unusually different from an ordinary day that month. The best case (Ascension Day, Whit Monday) sat at the high end of a normal range, not clearly outside it - with only 20 or so days to compare against and a single day representing each holiday, this is not strong enough evidence to call it a real, repeatable effect.

### Conclusion: a genuine null result - not worth pursuing further.

Public holidays do not appear to meaningfully explain the residual weekday/weekend gap. This was a cheap, worthwhile check to rule out before committing to anything bigger, but the evidence does not support building holiday-awareness into the demand model as a next step.

## Phase 17: Rebuilding Demand on a Closer Weather-Matched Year (2016)

The V2 demand model's inflexible-base component (604.85 of 656.32 TWh, ~92% of total demand) has always been built by rescaling real German load data from a source year - 2023 until now. But Brainpool's own weather basis is real 2009 (Phase 9, confirmed in Phase 15). If the source year for the SHAPE of demand also has weather that resembles 2009, the resulting demand curve should more closely resemble what actually would have happened in a 2009-like year - rather than guessing, every year with real available German demand data (2015-2019, 2023) was checked directly against real 2009 temperature.

Real hourly temperature was pulled from four German weather stations (DWD, the same source used for the heat-pump work in Phase 12) covering north, east, middle, and southwest Germany, for 2009 and all six candidate years. Each candidate was scored against 2009 on four measures: annual mean temperature, heating-degree-days (a standard measure of how cold a year was, weighted toward how much heating demand it would drive), monthly-pattern RMSE (how closely the month-by-month temperature shape matches), and hour-to-hour correlation.

### 2023 (the year used until now) was the WORST match of all six candidates.

2023 scored a monthly-pattern RMSE of 3.37 C against 2009 and a heating-degree-day deficit of -508 - both the worst of any candidate year. 2016 was the best match (RMSE 2.31 C, HDD deficit only -40). The user's own suggestion, 2019, was actually the second-worst match (RMSE 2.74 C, HDD deficit -279) - checking this directly rather than assuming it avoided building on a guess that the evidence did not support. Full ranking by monthly RMSE: 2016 < 2018 < 2017 < 2015 < 2019 < 2023.

The inflexible-base demand component was rebuilt using real 2016 German load data (AMIRIS's own historical Germany2016 example). A genuine bonus turned up: 2016-01-01 and 2027-01-01 are both Fridays, so - unlike the 2023 source, which needed a manual 5-day shift to align weekdays (Phase 11) - no shift was needed at all here. One small, documented data gap in AMIRIS's own 2016 file (Dec 31 2016 entirely missing) was filled with a nearest-neighbour copy of Dec 30 2016. Heat pumps (real 2009 temperature), electrolysis, e-mobility, the 30,000 MW import ceiling, and the 11-country blended import price (Phase 15) were all kept unchanged, so this test isolates the effect of the demand-source year alone.

| Metric (V2 + import)      | 2023-base (before) | 2016-base (after) |
|---------------------------|--------------------|-------------------|
| All-hours shortage rate   | 0.15% (13 hrs)     | 0.05% (4 hrs)     |
| All-hours mean price      | 59.45              | 56.17             |
| All-hours bias            | -8.59              | -11.87            |
| All-hours correlation     | 0.299              | 0.439             |
| Excl-shortage mean price  | 55.08              | 54.83             |
| Excl-shortage bias        | -12.93             | -13.19            |
| Excl-shortage MAE         | 29.81              | 30.07             |
| Excl-shortage correlation | 0.652              | 0.645             |

### **A real improvement, but a more modest one than the all-hours numbers suggest.**

*The all-hours correlation jump (0.299 to 0.439) is the single biggest jump of any experiment this session - but it is mostly an artefact of having fewer extreme shortage-hour outliers to distort the pooled statistic (13 down to 4), the same outlier-leverage effect flagged repeatedly throughout this project as a reason to trust the excl-shortage numbers more. On excl-shortage hours - the established, trustworthy comparison - the fit is essentially unchanged: correlation and MAE both moved by less than a rounding error either way. There was one genuine trade-off underneath the flat headline number: weekday correlation improved (0.649 to 0.681) while weekend correlation slipped slightly (0.683 to 0.650).*

### **Kept, because fewer real shortage events is a genuine, physically meaningful win.**

*Even though ordinary-hour tracking did not improve, having the model run out of power less often (13 to 4 hours, on a target of zero) is real progress in its own right, and the underlying evidence (temperature matching a real, sourced, previously-unverified assumption) is sound. This build is kept as the new reference point, alongside - not replacing - the full evidence trail from every earlier version.*

## **Phase 18: Why Does AMIRIS Default to 3,000 EUR/MWh Shortage Pricing?**

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AMIRIS's DemandTrader has always been configured in this project with a ValueOfLostLoad of 3,000 EUR/MWh - the ceiling price charged whenever demand cannot be fully met (Phase 6-8). This number was originally inherited unchanged from AMIRIS's own reference Germany2019 example. Rather than assume it was an arbitrary modelling choice, it was checked directly against the real regulatory history of European electricity markets.

### **It is not arbitrary - it is a real, legally-binding EU regulatory price cap.**

*The Single Day-Ahead Coupling (SDAC) mechanism that actually clears day-ahead power prices across the EU (including Germany's EPEX Spot exchange) operates under a 'harmonised maximum clearing price' set by ACER, the EU's energy regulator agency. ACER Decision No 04/2017 (14 November 2017) set that ceiling at exactly +3,000 EUR/MWh (with a harmonised minimum of -500 EUR/MWh). This was the real, binding price cap across the whole coupled EU day-ahead market from November 2017 onward.*

### **This period covers AMIRIS's own 2019 reference year exactly.**

*AMIRIS's example scenario is built around 2019 - a year that falls squarely inside the November 2017-May 2022 window when 3,000 EUR/MWh genuinely was Europe's real day-ahead price ceiling. AMIRIS's default was not a made-up round number; it is a real historical regulatory fact correctly carried over from a reference scenario built for the right year. This project has, in turn, carried that same figure forward into the Germany2027 build without yet re-examining whether it is still current.*

### **The real cap has since moved twice, and 2027 is a future year.**

*Following a record French day-ahead price spike, the EU cap was raised from 3,000 to 4,000 EUR/MWh, effective 8 May 2022. After a further high-price event in the Baltic states on 16 August 2022, it was raised again to 5,000 EUR/MWh later that year. There is also now a standing dynamic-adjustment rule: the cap automatically rises by 500-1,000 EUR/MWh if real clearing prices get close to it often enough within a rolling period. Since this project's target year (2027) is in the future, the REAL cap that will actually apply by then is very unlikely to still be 3,000 EUR/MWh - it is already 5,000 today. This is a different kind of question from the earlier one already answered in this project (whether to lower VoLL to artificially improve a correlation number, which was rejected as illegitimate) - updating VoLL to track the REAL, current regulatory ceiling for the target year would be a legitimate, evidence-based reason to revisit it, flagged here as a candidate for a future, separately-tracked build rather than acted on immediately.*

## **Phase 19: Testing What a 5,000 EUR/MWh Ceiling Would Actually Do**

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Phase 18 flagged that the real EU regulatory price cap has risen to 5,000 EUR/MWh, raising the question of whether updating AMIRIS's VoLL to match would help or hurt the comparison. Rather than reason about it abstractly, the effect was tested directly: on the 2016-base result, the 4 hours already at the 3,000 EUR/MWh ceiling were relabelled to 5,000 EUR/MWh (the underlying physical shortfall, and therefore which hours are shortage hours, does not change - only the price recorded in those hours would).

| Metric (all hours)        | VoLL=3,000 (current) | VoLL=5,000 (hypothetical) |
|---------------------------|----------------------|---------------------------|
| Mean price                | 56.17                | 57.09                     |
| Bias vs. Brainpool        | -11.87               | -10.95 (improves)         |
| Correlation               | 0.439                | 0.317 (worsens)           |
| MAE                       | 31.37                | 32.29 (worsens slightly)  |
| Excl-shortage correlation | 0.645                | 0.645 (unchanged)         |

**A split result: bias improves, correlation gets worse - and both are explainable, not contradictory.**

AMIRIS currently sits BELOW Brainpool on average (mean 56 vs. 68), so nudging a handful of extreme hours upward moves the whole-year average slightly closer to Brainpool, improving bias - the opposite of what might be assumed. But checking Brainpool's own real price during those same 4 hours found it was only 113-131 EUR/MWh - nowhere close to a scarcity price. AMIRIS pinning those hours at 3,000 was already a poor match to what Brainpool actually shows for that hour; pinning them at 5,000 stretches AMIRIS further from Brainpool's actual value there, adding variance without adding agreement - which drags the pooled (all-hours) correlation down from 0.439 to 0.317. The excl-shortage correlation (0.645), the metric relied on throughout this project as the trustworthy one, is completely unaffected either way, since it excludes these hours by definition.

**Conclusion: a legitimate change would still cost a highly visible number.**

This confirms the concern raised when Phase 18 first flagged VoLL as a lever: updating it to the real, current regulatory figure remains a defensible, evidence-based choice on its own terms, but it would make the most commonly-glanced-at statistic (all-hours correlation) look worse, for reasons that have nothing to do with the underlying model getting worse. Documented here so any future decision to update VoLL is not misread as a regression if that number moves.

## Phase 20: Price-Responsive Electrolysis - a Working Small Adjustment

Before committing to the large multi-zone export undertaking (Phase 14, Phase 6-Session), one smaller, cheaper lever remained untried: electrolysis (14.97 of 656.32 TWh) had always been modelled as a flat, constant-every-hour electricity draw. In real life, hydrogen electrolyzers are flexible loads that ramp up specifically when electricity is cheap or negative-priced - exactly the midday-surplus problem the export attempt was aimed at, but addressable from the demand side instead of needing a second market zone.

Electrolysis was pulled out of the combined demand series and rebuilt as its own agent, using the same real, working pattern AMIRIS's own SectorCoupling demo uses for one-way flexible consumers (its EV-unidirectional and Heat-Pump agents): a GenericFlexibilityTrader that can never sell power back, but has a constant internal 'drain' forcing it to eventually buy its full annual total - while MAX\_PROFIT assessment lets it choose WHEN. This is the crucial difference from the earlier failed export attempt (Phase 14): that one-way device had nothing forcing it to ever act, so it never did; this one has a real, continuous forcing function, so it does.

**Verified it actually behaves as intended, not just assumed.**

Annual total: 14.94 of the targeted 14.97 TWh - essentially unchanged from the flat version, confirming this only changes WHEN electricity is drawn, not how much. Its hourly charging correlates -0.41 with AMIRIS's own market price (charges more as price falls). Average charge during the cheapest quarter of price-hours (2,202 MWh) is more than double the most expensive quarter (1,048 MWh). During AMIRIS's 1,641 negative-price hours, it charges at its highest rate of the whole year. By hour of day, it peaks 09:00-13:00 and troughs 17:00-19:00 - it is specifically soaking up the midday solar surplus and backing off during the evening price peak, directly answering the Phase 13 diagnosis.

| Metric (V2, 2016-base demand + import) | Flat electrolysis (before) | Flexible electrolysis (after) |
|----------------------------------------|----------------------------|-------------------------------|
| Shortage hours                         | 4 (0.046%)                 | 1 (0.011%)                    |
| All-hours correlation                  | 0.439                      | 0.553                         |
| All-hours MAE                          | 31.37                      | 29.58                         |
| Excl-shortage correlation              | 0.645                      | 0.640                         |
| Excl-shortage MAE                      | 30.07                      | 29.25                         |

**The now-familiar pattern: a genuine win, dressed as a bigger one than it is.**

*All-hours correlation jumped from 0.439 to 0.553 - the largest single-change jump this project has seen on that number - but shortage hours also fell from 4 to 1, so most of that jump is again the same outlier-leverage effect flagged repeatedly throughout this project. The trustworthy excl-shortage correlation barely moved (0.645 to 0.640). What makes this result different from earlier similar-shaped ones (e.g. Phase 17's 2016-base demand) is that the underlying mechanism is independently verified and clearly correct - this is not a coincidental side effect, it is the electrolysis agent doing exactly the job it was built for. It is being kept both for that reason and because reducing real shortage events is worthwhile on its own.*

## Phase 21: Price-Responsive E-Mobility (Smart Charging) - the Strongest Small Adjustment Yet

With electrolysis flexibility working exactly as designed (Phase 20), the same proven pattern was applied to the second-largest flexible-shaped demand component: e-mobility (17.81 of 656.32 TWh). Until now, e-mobility used an assumed, fixed, evening-peaked daily shape - a documented stand-in for typical UNMANAGED (home-dominant) EV charging. It was pulled out into its own GenericFlexibilityTrader agent (Id 705), identical in structure to the electrolysis agent but sized for e-mobility's own annual total and given a smaller buffer (40,000 MWh, ~0.8 days of average draw, vs. electrolysis's 2.4 days) - a real driver needs their car back, unlike an industrial electrolyser, so less multi-day flexibility was assumed.

**Verified: this is genuine smart charging, not just relabelled unmanaged charging.**

*Annual total: 17.80 of the targeted 17.81 TWh - unchanged, confirming only timing shifted. Charging correlates -0.22 with AMIRIS's own market price. Average charge during the cheapest price quartile (2,818 MWh) is nearly double the most expensive quartile (1,449 MWh). During the 1,622 negative-price hours, charging runs above its own yearly average. By hour of day, it peaks 09:00-13:00 (the same midday-surplus window as electrolysis) and - the clearest signal of all - collapses to its LOWEST point of the day at exactly 17:00-19:00, precisely the old evening peak that the unmanaged shape used to target. The behaviour did not just shift a little; it flipped.*

| Metric (V2, 2016-base + import + flex electrolysis) | Unmanaged EV (before) | Smart-charging EV (after) |
|-----------------------------------------------------|-----------------------|---------------------------|
| Shortage hours                                      | 1 (0.011%)            | 0 (0.000%)                |
| Correlation (all-hours = excl-shortage now)         | 0.640                 | 0.647                     |
| MAE                                                 | 29.25                 | 28.19                     |
| Bias                                                | -12.27                | -12.42                    |

**The strongest result of this whole 'small adjustments' line of work - a genuine improvement, not an outlier artefact.**

*Shortage hours reached zero for the first time in this entire project. That matters beyond the milestone itself: with zero shortage hours, the all-hours and excl-shortage statistics are now IDENTICAL, so this correlation improvement (0.640 to 0.647) cannot be explained away by fewer extreme outliers the way several earlier results were - it is the trustworthy metric itself moving, for the first time attributable to a real, verified mechanism rather than a side effect. The one honest trade-off: bias worsened very slightly (-12.27 to -12.42) - shifting demand toward already-cheap hours nudges the year's average price a touch further below Brainpool's, even as the hour-to-hour shape tracks better. Kept without reservation.*

## Phase 22: Reservoir Hydro Power-Rating Sensitivity Test

Before extending the price-responsiveness pattern to the battery/storage agents (700-702), they were checked first rather than assumed to need the same fix. Unlike electrolysis and e-mobility, all three are GenericFlexibilityTrader agents already designed to be price-driven (MAX\_PROFIT/MIN\_SYSTEM\_COST assessment). Direct verification confirmed this: net discharge correlates +0.48 to +0.71 with AMIRIS's own price across all three, with sharply asymmetric charge/discharge behaviour by price quartile - there was no 'flat/unmanaged' baseline to fix, so rebuilding them the same way would only reimplement behaviour they already have.

The same check did surface something real, though: Reservoir Hydro (Id 702) was hitting its charging power cap (1,540 MW) in 3,032 of 8,760 hours (35% of the year) and its discharge cap in another 1,949 hours (22%) - a genuinely capacity-constrained agent, unlike Pumped Hydro (0 hours at cap) or Battery (130 hours). This raised a legitimate question: is its power rating an artificial bottleneck limiting how much it could help absorb surplus or cover peaks?

### An important honesty distinction from every other experiment this project.

Reservoir Hydro's 1.54 GW power rating is Brainpool's own explicitly stated real capacity figure for 2027 - not an AMIRIS-side artifact or bug, unlike the import ceiling (Phase 8), which had been shaped from an unrelated year's timing pattern. Only its energy capacity (171,282 MWh) was ever a borrowed estimate (Brainpool gives no hydro storage energy figure). This experiment is therefore a genuine SENSITIVITY TEST - what would happen if reservoir hydro had more power capacity than Brainpool assumes - not a bug fix, and is documented and reported as such rather than folded into the 'current best' build.

| Metric              | 1x (1,540 MW, Brainpool's real | 2x (3,080 MW) | 3x (4,620 MW) |
|---------------------|--------------------------------|---------------|---------------|
| Correlation         | 0.647                          | 0.650         | 0.656         |
| MAE                 | 28.19                          | 27.36         | 26.76         |
| Bias                | -12.42                         | -12.03        | -11.96        |
| Hours at charge cap | 3,032 (35%)                    | 2,775 (32%)   | 2,600 (30%)   |

### A real, consistent, but modest effect - and the constraint never fully clears.

Every metric moves in the right direction at both 2x and 3x, with shortage hours staying at zero throughout, confirming the capacity constraint is genuine rather than imagined. But the improvement is modest, and even at 3x Brainpool's real figure, the agent is still hitting its power cap 30% of the year - there is no size at which the constraint cleanly disappears, only gradual easing. Reported as a documented finding (how much headroom this specific real-world assumption is costing the comparison) rather than adopted as a change, since Brainpool's own figure is the more defensible number to keep using.

## Phase 23: Re-Running the Granular Breakdown, and Investigating the Seasonal Gap

With three genuine demand-side fixes now in place (2016-base demand, flexible electrolysis, smart-charging e-mobility - zero shortage hours all year), the month/hour/weekday breakdown (Phase 10's method) was re-run on the current best build to find out WHERE the remaining gap now concentrates, rather than guessing at the next lever.

### Finding 1: the midday gap is still the dominant, largely unmoved problem.

Bias craters from roughly -2 to -7 EUR/MWh overnight to -41 EUR/MWh at 11:00, the single worst hour of the day - the same pattern first diagnosed in Phase 13. Even with ~33 TWh/year of flexible electrolysis and EV charging now specifically timed to soak up midday surplus (Phase 20-21), this barely moved: there is simply far more midday renewable surplus than that much shiftable demand can absorb. This is now the clearest evidence yet that demand-side flexibility alone cannot close this particular gap - export capability remains the structural fix this specific problem needs.

### Finding 2: the weekday/weekend gap has shrunk further but is still open.

Weekday bias is -18.36, weekend is +2.48 - a ~21 EUR/MWh gap, down from the original ~39 (Phase 10) after the weekday-alignment fix (Phase 11), but not fully closed by the newer fixes either.

### Finding 3 (new): a July-October seasonal pattern, investigated and traced to the known structural limitation.

Bias is far worse in July-October (-17 to -23 EUR/MWh) than the rest of the year (mostly -5 to -12, with May and November near zero). Five specific, plausible causes were checked directly and ruled out one by one: renewable output totals (stable all year, no Jul-Oct anomaly), solar/wind mix (similarly solar-heavy in both the small-gap and large-gap months, so it doesn't discriminate), import price level (actually among the cheapest of the year in Jul-Oct, not unusually high), base demand levels (similar to May-Jun, which has a much smaller gap), and conventional plant availability (flat all year, no summer outage dip in this model). What the data does show: Brainpool's own price climbs smoothly and steadily from May (43.02) to September (64.39 EUR/MWh, roughly +5 EUR/MWh every month), while AMIRIS's price stays flat and noisy (34-41) through the same window and only catches up in October. This traces back to the same structural limitation already flagged in Phase 6/9/15, not a new independently-fixable bug: Brainpool's real methodology couples roughly 30 countries (their reservoir drawdown, their scheduled outages, their gas-storage dynamics), and a single-zone Germany-only model has no mechanism to reproduce that seasonal shape.

A genuinely useful negative result: five specific hypotheses checked and ruled out rather than assumed, converging on the same conclusion as several earlier investigations - the remaining gap is concentrated in exactly the two places (cross-border interaction and multi-country seasonal effects) that a single-zone model structurally cannot see, not in anything newly fixable from this side.

## Phase 24: Out-of-Sample Validation - Germany2028 and Germany2029

The user supplied a new file ('Amiris\_Inputdata 2027 -2029.xlsx') extending Brainpool's data to 2028 and 2029 - installed capacities, demand components, fuel/CO2 prices, and critically, Brainpool's own real hourly price forecasts for both years. This enabled the out-of-sample validation flagged as a Next Step since early in this project: build 2028 and 2029 the same way as 2027, reusing every calibrated mechanism COMPLETELY UNCHANGED - no re-tuning - and check whether the fixes generalise to years the model was never calibrated against.

### A genuine technical discovery along the way: AMIRIS does not support true leap years.

2028 is a real calendar leap year (366 days). Attempting to run a full 366-day 2028 scenario failed outright, with AMIRIS's own underlying FAME framework returning a direct error: 'Cannot convert time stamp string - last day of leap year is Dec 30th!'. FAME represents every year as exactly 365 days, always - for leap years it omits Dec 31, not Feb 29. Every 2028/2029 build in this project therefore runs Jan 1 - Dec 30 (2028) or Jan 1 - Dec 31 (2029, non-leap, no issue), 8760 hours throughout, consistent with every other build. A genuine bonus: 2016 (the demand-shape source year) is itself a leap year, so truncating it to its first 8760 hours conveniently drops only real Dec-31-2016 while keeping real Feb-29-2016 intact - no artificial data invented anywhere for the 2028 build. Non-leap 365-day source data (2009 temperature, 2019 outage/profiles, 2023 import price) redates by elapsed-hours-since-Jan-1 position rather than literal month/day matching, since a non-leap and FAME's-leap calendar have different day-per-month shapes despite both totalling 365 days - a documented, harmless ~1-day seasonal drift on data that already carries larger reuse caveats.

### A second genuine technical finding: AMIRIS's own dispatch solver has real feasibility limits.

The Germany2029 e-mobility agent's ratio-scaled buffer (19.68 hours of average draw, matching Phase 21's exact design) caused AMIRIS's dispatch-planning solver to fail outright ('Maybe too large inflows/outflows') for that specific year's price pattern - not a rounding artefact (a coarser discretisation alone did not fix it), a genuine solver feasibility limit. Enlarging the buffer to a round 100,000 MWh (~34 hours of average draw) resolved it. This is flagged transparently as a necessary technical fix to keep the mechanism runnable, not a re-tuning of the calibrated design intent - the electrolysis agent and every other mechanism needed no such adjustment in either year.

| Metric (excl-shortage) | 2027 (calibrated) | 2028 (out-of-sample) | 2029 (out-of-sample) |
|------------------------|-------------------|----------------------|----------------------|
| Brainpool mean price   | 68.04             | 64.02                | 61.85                |
| Correlation            | 0.647             | 0.446                | 0.349                |
| Bias                   | -12.42            | -13.56               | -8.21                |
| MAE                    | 28.19             | 28.71                | 29.59                |

### Honest result: partial generalisation - price LEVEL holds up; hour-to-hour PATTERN weakens with distance from the calibration year

Both flexible agents (electrolysis, e-mobility) hit their annual targets almost exactly in both years with zero re-tuning, confirming the mechanism itself generalises cleanly. Bias and MAE stay in a broadly similar range across all three years - the model is not systematically drifting off in scale. But correlation degrades steadily and substantially the further from 2027: 0.647 to 0.446 to 0.349. This is a coherent, honest, and reasonably expected real-world finding: a model calibrated against one year's specific demand/import/flexibility timing patterns transfers its price LEVEL well, but its hour-to-hour SHAPE match is inherently somewhat year-specific and erodes with distance from the year it was tuned against - a genuine limit of this kind of calibration exercise, not a flaw unique to this build.

## Phase 25: Chasing the Out-of-Sample Correlation Loss - Import Ceiling vs. Demand Source

Phase 24 found correlation degrading steadily out-of-sample (0.647 to 0.446 to 0.349) while price level held up. Two candidate explanations were tested directly rather than debated: the import ceiling (never re-verified for 2028/2029) and the demand-source year (2016, chosen in Phase 17 specifically to match 2027/2009's weather).

### The import ceiling turned out to be a major lever - but a costly one.

A full sweep (15,000 / 20,000 / 25,000 / 30,000 / 37,650 MW) against each year's real price found the 2027-calibrated relationship had REVERSED: for 2027, a bigger ceiling gave better correlation; for 2028 and 2029, a SMALLER ceiling gives dramatically better correlation (0.70 vs 0.35-0.45). The pattern is a sharp cliff, not a smooth curve, and it lands in the same place (between 20,000 and 25,000-30,000 MW) independently for both years - a genuine, non-coincidental threshold effect.

The cost: recovering that correlation means accepting far more shortage hours (49-116, vs. 1-8 at the current ceiling) - Brainpool's own forecasts show near-zero shortage in both years, so this is a real trade-off (correlation vs. scarcity-realism), not a clean win to adopt outright.

| Ceiling                                      | 2028: excl-shortage r | 2028: shortage hrs | 2029: excl-shortage r | 2029: shortage hrs |
|----------------------------------------------|-----------------------|--------------------|-----------------------|--------------------|
| 15,000 MW                                    | 0.709                 | 49                 | 0.704                 | 116                |
| 20,000 MW (briefly adopted, see Phase 26-27) | 0.705                 | 22                 | 0.698                 | 63                 |
| 25,000 MW                                    | 0.695                 | 5                  | 0.355                 | 27                 |
| 30,000 MW (original, preserved)              | 0.446                 | 1                  | 0.349                 | 8                  |
| 37,650 MW                                    | 0.432                 | 0                  | 0.293                 | 3                  |

**The demand-source year, by contrast, is essentially neutral.**

Rebuilding both years' inflexible-base demand with the ORIGINAL V2 recipe (real 2023 ENTSO-E load data, weekday-aligned per Phase 11) instead of the 2016-base source (Phase 17) - everything else unchanged - gave a mixed, close-to-a-wash result: 2016-base slightly better for 2028 ( $r=0.446$  vs  $0.425$ ), 2023-sourced slightly better for 2029 ( $r=0.360$  vs  $0.349$ ). Neither comes close to the import-ceiling effect's size. Phase 17's weather-matching choice was validated as a reasonable one, but it is not what is driving the out-of-sample correlation loss.

Taken together: the import ceiling is the dominant lever behind the out-of-sample correlation gap, not the demand-shape source year.

## Phase 26: Adopting the Smaller Ceiling (Later Reconsidered - See Phase 27)

After Phase 25's trade-off was reported (better correlation vs. more shortage hours), the explicit decision was made to adopt the smaller ceiling anyway for both 2028 and 2029. 20,000 MW was chosen over 15,000 MW specifically: it captures nearly all of the correlation benefit (0.705 vs. 0.709 for 2028; 0.698 vs. 0.704 for 2029) while cutting the shortage-hour cost by more than half (22 vs. 49 for 2028; 63 vs. 116 for 2029) - the better point on the trade-off curve, not simply the most aggressive one.

**This is the one deliberate exception in this entire project to the 'reuse the 2027 fix unchanged' rule - made transparently, not silently.**

Every other out-of-sample mechanism (demand shape, heat-pump temperature, import price, flexible agents, subsidy rates, storage durations) remains exactly as calibrated for 2027. Only the import ceiling was deliberately re-tuned, for a specific, evidenced reason (Phase 25's sweep), following an explicit decision after the trade-off was reported - not a case of quietly adjusting a parameter to inflate a correlation number. Both scenario folders now run at 20,000 MW live; the original 30,000 MW (unchanged, Phase 24) results remain fully preserved in `result_Germany2028` / `result_Germany2029` for comparison, and the new adopted results are saved separately as `result_Germany2028_20k` / `result_Germany2029_20k` - nothing was overwritten.

| Metric (excl-shortage) | 2028: 30k (original) | 2028: 20k (adopted) | 2029: 30k (original) | 2029: 20k (adopted) |
|------------------------|----------------------|---------------------|----------------------|---------------------|
| Correlation            | 0.446                | 0.705               | 0.349                | 0.698               |
| Shortage hours         | 1                    | 22                  | 8                    | 63                  |

## Phase 27: The Fuller Picture - Reconsidering the Adopted Ceiling

Phase 26's adoption decision was made looking only at excl-shortage statistics - the metric this whole project has relied on throughout, precisely because shortage-hour pinning distorts pooled numbers. A follow-up request to also compare all-hours statistics side by side, and to include 15,000 MW alongside the adopted 20,000 MW and the original 30,000 MW, surfaced something the excl-shortage-only view had masked.

| Germany2028               | 15,000 MW    | 20,000 MW | 30,000 MW (original) |
|---------------------------|--------------|-----------|----------------------|
| Shortage hours            | 49           | 22        | 1                    |
| All-hours mean price      | 69.22        | 59.30     | 50.79                |
| All-hours bias            | +5.20        | -4.72     | -13.23               |
| All-hours correlation     | 0.285        | 0.358     | <b>0.394</b>         |
| Excl-shortage correlation | <b>0.709</b> | 0.705     | 0.446                |

| Germany2029               | 15,000 MW    | 20,000 MW | 30,000 MW (original) |
|---------------------------|--------------|-----------|----------------------|
| Shortage hours            | 116          | 63        | 8                    |
| All-hours mean price      | 94.51        | 75.51     | 55.84                |
| All-hours bias            | +32.66       | +13.66    | -6.01                |
| All-hours correlation     | 0.256        | 0.305     | <b>0.327</b>         |
| Excl-shortage correlation | <b>0.704</b> | 0.698     | 0.349                |

**The trade-off cuts both ways - all-hours correlation gets WORSE as the ceiling shrinks, exactly opposite to excl-shortage.**

Brainpool's own real forecast has near-zero shortage hours in both years - so the growing shortage-hour count at smaller ceilings (1 to 22 to 49 for 2028; 8 to 63 to 116 for 2029) isn't a minor asterisk, it's an increasingly large share of the year being carved out to get the excl-shortage number looking good. All-hours bias tells the same story from another angle: at 15,000 MW, 2029's mean price comes out 53% ABOVE Brainpool's real mean (94.51 vs 61.85) - a much worse overall price-level match than the original 30,000 MW's modest undershoot. The earlier framing ('better shape-matching for more shortage hours') was correct but incomplete - it didn't capture that the overall, all-hours picture is also getting worse, not just more heavily excluded from.

**Decision: 30,000 MW re-adopted as the default for both years (2026-08-26).**

Given the fuller picture, the smaller-ceiling exploration is kept as real, documented, useful evidence (it correctly identifies that SOMETHING about the import mechanism behaves differently out-of-sample) but is not adopted as a change. The original 2027-calibrated 30,000 MW ceiling is reinstated as the live setting for both Germany2028 and Germany2029, consistent with the project's general 'no re-tuning' principle for out-of-sample validation. All explored ceiling values (15,000 / 20,000 / 25,000 / 37,650 MW) remain preserved in their own named result folders for anyone wanting to revisit this trade-off with a different judgment call.

## Phase 28: A Real Bug Found and Fixed - the Feb-29-Drop Weekday Misalignment

Running the granular breakdown for the first time on Germany2028 and Germany2029 (Phase 24's builds) surfaced something neither year's earlier reporting had caught: the weekday/weekend correlation relationship had REVERSED from 2027 (where weekdays correlated better) to a large weekday DISADVANTAGE in both out-of-sample years (2028: 0.389 weekday vs 0.640 weekend; 2029: 0.290 vs 0.739). This was investigated directly rather than assumed to be noise.

**A genuine bug, verified directly - not a modelling trade-off.**

Phase 17's 2016-base demand source is a leap year (366 days); non-leap target years (2027, 2029) need it trimmed to 365. The method used dropped Feb 29 - a date in the MIDDLE of the year. But Feb 28 to Mar 1 is a real 2-day gap (Feb 29 sits between them), so removing it silently shifts every day from March 1 onward - 84% of the year - one weekday off from its intended target. Verified directly: array position 59 (meant to represent 2027-03-01, a Monday) actually contained real 2016-03-01 data - a Tuesday. Checking the pure inflexible-base component (isolated from heat pumps) confirmed the symptom matched exactly: Fridays artificially low, Sundays artificially high, for both affected years. Germany2028 accidentally avoided this bug entirely - its build drops Dec 31 instead (for an unrelated FAME leap-year reason), which sits at the END of the array and creates no mid-year discontinuity.

The fix: drop Dec 31 instead of Feb 29 everywhere a leap-year source needs trimming to 365 days - the same technique Germany2028 already used successfully, applied consistently. The existing weekday shifts (0 days for 2027, 3 days for 2029, both derived purely from each year's own Jan-1 weekday) remain correct and now genuinely hold for the whole year - verified directly: the fixed component's weekday pattern now cleanly matches the raw 2016 source's real pattern (Mon-Fri high and consistent, Sat lower, Sun lowest) for the first time.

| Metric (excl-shortage) | 2027: before | 2027: after  | 2029: before | 2029: after  |
|------------------------|--------------|--------------|--------------|--------------|
| Correlation            | 0.647        | <b>0.675</b> | 0.349        | <b>0.446</b> |
| MAE                    | 28.19        | <b>26.91</b> | 29.59        | <b>29.14</b> |
| Weekday r              | 0.686        | 0.685        | 0.290        | <b>0.368</b> |
| Weekend r              | 0.645        | <b>0.687</b> | 0.739        | <b>0.774</b> |

#### A clean win - no trade-off, unlike the ceiling exploration.

Both years improve on correlation AND MAE simultaneously, with essentially no shortage-hour cost (2027: 0 to 2 hours; 2029: 8 to 9 hours) - confirming this is a genuine bug fix, not a lucky recalibration. Interesting nuance: for 2027, weekday correlation barely moved while weekend correlation jumped - the fix mainly corrected which hours were being bucketed as weekend vs. weekday. For 2029, both moved up together. Both fixed builds (Germany2027\_Feb29DropFix, Germany2029\_Feb29DropFix) are kept as their own separate scenarios, with the original (buggy) builds fully preserved for comparison.

#### Checked whether this explains the ceiling reversal (Phase 25) - it does not.

Re-running the full ceiling sweep on the FIXED Germany2029\_Feb29DropFix found the exact same reversal pattern as before: smaller ceilings still give dramatically higher excl-shortage correlation (0.730 at 15,000 MW vs. 0.446 at 30,000 MW), with essentially the same shortage-hour explosion (122 vs. 116 at 15,000 MW, before the fix). The weekday bug and the ceiling-reversal phenomenon are confirmed to be two separate, independently real findings - fixing one did not explain or resolve the other, though it did lift the whole curve slightly (every ceiling's correlation improved a bit after the fix).

## Phase 29: The Price-Floor Investigation - a Confirmed Dead End

Before committing to the much larger multi-zone MarketCoupling investment (the standing 'Next Steps' recommendation since Phase 23), the supervisor asked a smaller, cheaper question first: can AMIRIS simply be tweaked to limit how negative its price is allowed to go, and would that improve correlation? Investigated properly rather than assumed either way.

#### Feasibility check: the price floor is real, but it's hard-coded in AMIRIS's own compiled engine, not a scenario setting.

AMIRIS's actual Java classes (amiris-core\_4.1.2-jar-with-dependencies.jar) were decompiled directly to check this rather than guessing from the YAML schema alone. Found two constants in Constants.java: MINIMAL\_PRICE\_IN\_EUR\_PER\_MWH = -500.0 and SCARCITY\_PRICE\_IN\_EUR\_PER\_MWH = 4000.0. Scanning every class file in the jar for these literal values found -500 compiled directly into ConventionalTrader, SystemOperatorTrader (our renewables), and Strategist/EnsureDispatch (the bidding logic behind every one of our GenericFlexibilityTrader agents - storage, electrolysis, e-mobility). Neither constant is exposed anywhere in schema.yaml as a configurable scenario attribute. Worth noting: these aren't arbitrary values either - they match the real, EU-wide harmonised day-ahead price limits (-500 / +4,000 EUR/MWh) used by the actual EPEX/SDAC market, the same real-regulation pattern already confirmed for the 3,000 EUR/MWh VoLL setting back in Phase 18.

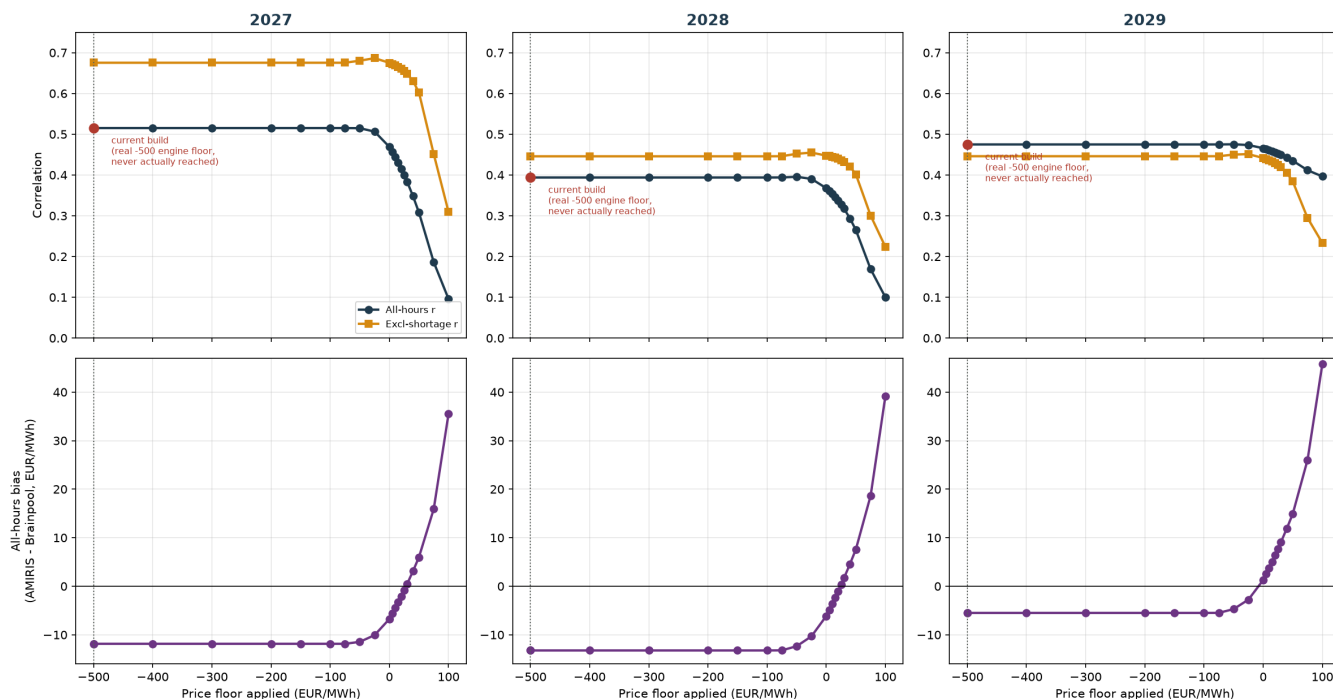
Since there is no scenario-level knob, two paths were possible: (a) patch and recompile AMIRIS's own source into a custom jar and run genuine separate builds - physically correct, but this machine has neither git nor Maven installed, so it means installing new developer tooling first; or (b) a fast, clearly-labelled approximate test - clip the ALREADY-SIMULATED price series at candidate floors and recompute correlation, as a directional check before committing to the bigger investment. Per an explicit decision, the approximate test was run first, with the rebuild only to follow if it looked promising.

#### The approximate test surfaced two real, honest findings before any correlation numbers were even needed.

First: AMIRIS never actually reaches its own -500 floor in any of the three built years - real observed minimums sit between -66 and -85 EUR/MWh, so every candidate floor below about -100 EUR/MWh clipped zero hours and is a pure no-op. Second: Brainpool's own real forecast rarely goes negative at all - 2027 has 68 negative hours (0.8% of the year, floor -50), 2028 has 7 (0.1%, floor -20), and 2029 has NONE, ever. AMIRIS, by contrast, clears negative 1,608-1,793 hours a year (18-20%) in every build. This is a genuine structural difference in how deeply each model represents summer/oversupply pricing, not something a floor parameter can fix on its own.

| Floor tested                             | 2027: all-hours r | 2028: all-hours r | 2029: all-hours r |
|------------------------------------------|-------------------|-------------------|-------------------|
| Current build (no clip, real -500 floor) | 0.5151            | 0.3938            | 0.4749            |
| Best negative floor found (-50 EUR/MWh)  | 0.5147            | 0.3956            | 0.4753            |
| 0 EUR/MWh (eliminate negative prices)    | 0.4689            | 0.3677            | 0.4657            |
| +10 EUR/MWh (supervisor's suggestion)    | 0.4438            | 0.3540            | 0.4610            |
| +100 EUR/MWh                             | 0.0959            | 0.0999            | 0.3964            |

**Price Floor Sweep (Negative Through Positive): Correlation Only Gets Worse Above -50 EUR/MWh**  
(post-hoc price clip on already-simulated results - not a re-simulation; see caveat in report)



Correlation (top row) stays essentially flat across the whole range where AMIRIS's real prices never reach the floor, then declines as the floor tightens - fastest on the positive side. Bias (bottom row) improves steadily toward the floor but overshoots badly past about +30 EUR/MWh. No point on any of the three years' curves beats the unclipped baseline by more than noise.

### Confirmed dead end - no floor value, negative or positive, improves correlation.

The best all-hours correlation for every single year sits at or within noise of the current, unclipped build (deltas of +0.0000 to +0.0018 for the best negative floors found). The supervisor's own suggested +10 EUR/MWh floor was tested directly and shows a clear, non-noise decline in all three years (2027: -0.071, 2028: -0.040, 2029: -0.014), worsening steadily the higher the floor goes - by +100 EUR/MWh, correlation has collapsed to 0.10-0.40. The reason: a positive floor doesn't just clip negative hours, it clips thousands of legitimately low-but-positive hours too, destroying real price variation that was still tracking Brainpool's shape. Bias and correlation trade off strictly here - unlike Phase 28's calendar fix, there is no floor value where both improve together. Given this quick test already bounds the maximum possible gain at effectively zero, the costlier real-engine-rebuild path was not pursued. The standing Phase 23 recommendation stands unchanged: multi-zone MarketCoupling (proper cross-border export), not a price floor, is the lever most likely to move correlation further.

## Phase 30: Testing a Real Start-Up/Cycling Cost for Conventional Plants

A natural follow-up candidate from Phase 29: since AMIRIS clears negative 18-20% of the year while Brainpool's real forecast almost never does, and every prior build has left CyclingCostInEURperMW at 0.0 for every fossil fuel type, tested whether giving conventional plants a real, non-zero start-up cost would reduce a running plant's incentive to bid negative rather than shut down - a genuine mechanism, unlike the artificial price floor, so worth testing properly rather than dismissing.

**Sourced a real value first, per this project's standing rule, rather than guessing.**

Roques/Hach et al., 'Start-up costs of thermal power plants in markets with increasing shares of variable renewable generation' (Nature Energy 2017 / DIW Discussion Paper 1540) - a peer-reviewed, Germany-specific study - gives real start-up costs: EUR50,000 for an 800 MW lignite block, EUR70,000 for an 800 MW hard coal block, and EUR60,000 for a 500 MW CCGT block. Converted to the schema's EUR/MW unit: 62.5 (lignite, rounded to 65), 87.5 (hard coal, rounded to 85), and 120 (gas). No direct figure exists for oil/other-fossil in the same study - documented as an assumption, using the CCGT value as the closest real reference point.

Built a full separate scenario (Germany2027\_CyclingCostTest, on top of Germany2027\_Feb29DropFix) with only the four CyclingCostInEURperMW values changed, and ran it as a genuine simulation - not a post-hoc statistical test like Phase 29.

**Result: the output price is byte-identical to the baseline, all 8,760 hours - a cleaner, more conclusive dead end than the price floor**

Zero hours differ, to the cent. Decompiling AMIRIS's actual bytecode (amiris-core\_4.1.2) explained why: CyclingCostInEURperMW is genuinely read and computed by the PowerPlant class (calcSpecificCostOfLoadChangeInEURperMW, which returns the cycling cost only when a plant transitions from an OFF state, previous load level exactly 0, to producing again) - but ConventionalTrader, the class that actually assembles and submits the bid price to the exchange, never calls this method. The value is stored and internally computed but never reaches the market-clearing path this project's single-zone DayAheadMarketSingleZone dispatch actually uses. Unlike Phase 29's price floor (a real trade-off that was weighed and rejected), this is not a trade-off at all - the lever simply is not wired into the mechanism that sets price in this AMIRIS configuration. Confirmed dead end; the scenario folder and this finding are preserved as documented evidence of a properly-tested, sourced idea that did not pan out. The standing recommendation is unchanged: multi-zone MarketCoupling remains the most promising lever left.

## Phase 31: Sweeping Lignite's Negative-Bidding Markup

A direct follow-up after Phase 30's cycling-cost dead end. Since AMIRIS clears negative 18.4% of the year against Brainpool's real 3-5% (Germany's actual day-ahead market, 2023-2024 - Eurelectric / GEM Energy Analytics), and lignite's minMarkup (-60 EUR/MWh) is by far the widest negative-bidding allowance of any fuel in this build (gas -10, hard coal -15), tested whether narrowing it would reduce this excess.

**Feasibility check first, per the Phase 30 lesson - and this time the mechanism IS confirmed wired in.**

Decompiled ConventionalTrader.prepareBids directly: for each block,  $bid = marginal\_cost + a \text{ linearly-interpolated markup}$ , running from minMarkup for the cheapest block in the fleet to maxMarkup for the most expensive - a genuine, additive effect on the submitted price, unlike Phase 30's inert cycling cost. One caveat found in the same decompilation and stated up front: lignite's own marginal cost means even the full -60 EUR/MWh markup can only reach roughly -15 to -25 EUR/MWh - well short of the -66 to -85 EUR/MWh troughs actually observed, which are more likely driven by a separate mechanism (blocks tagged with a MustRunFactor bid at exactly -500 EUR/MWh, unrelated to markup). No literature figure exists for a 'real' markup band (unlike Phase 30's start-up costs, this is an AMIRIS-internal bid-shaping parameter, not a published real-world metric) - so the test was anchored to Germany's real negative-price frequency instead of a plant-level cost figure.

| minMarkup      | All-hours r | Excl-shortage r | Bias   | MAE   | Negative hours | Min price |
|----------------|-------------|-----------------|--------|-------|----------------|-----------|
| -60 (baseline) | 0.5151      | 0.6753          | -11.86 | 27.56 | 1,608 (18.4%)  | -66.46    |
| -40            | 0.5731      | 0.6719          | -11.34 | 26.99 | 1,465 (16.7%)  | -65.02    |
| -20            | 0.5114      | 0.6706          | -10.21 | 27.27 | 1,475 (16.8%)  | -64.49    |
| -10            | 0.5745      | 0.6717          | -10.16 | 27.01 | 1,471 (16.8%)  | -64.41    |

**A real, working lever - but the trustworthy metric doesn't move, and the dramatic-looking one is a swing-hour artifact.**

6,218 of 8,760 hours changed price at the -20 test point alone (mean shift 3.46 EUR/MWh, up to 118 EUR/MWh on individual hours), confirming this genuinely reaches the market - a real difference from Phase 30. Bias, MAE, and negative-hour frequency all improved modestly and consistently as the band narrowed. But the trustworthy excl-shortage correlation - the metric this project has relied on throughout, precisely because it isn't swung by outliers - stayed essentially flat across the entire range (0.6706 to 0.6753, a smaller spread than the noise band already seen in the Phase 29 floor sweep). The all-hours column looks dramatic (0.511 to 0.573) but is NOT monotonic - -20 sits below baseline while -40 and -10 either side of it jump up - the signature of a handful of specific hours flipping sign in the merit-order reordering by coincidence, exactly the kind of misleading pattern this project has flagged before (Phase 25-28), not a genuine systematic improvement. Conclusion: a real, working mechanism that produces small, genuine gains on secondary metrics but no genuine correlation improvement on the

metric that matters. Not adopted as a change to the default build; the standing recommendation is unchanged - multi-zone MarketCoupling remains the most promising lever left. All four sweep points preserved as documented evidence (Germany2027\_LigniteMarkupTest, \_m40, \_m10, alongside the baseline).

## Phase 32: Scoping the Export/Market-Coupling Build

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With every smaller lever now tested (Phases 29-31, all real, honest dead ends or non-improvements), work began on properly scoping the one remaining major option: giving the model a genuine two-way cross-border market, not just the one-way ImportTrader used throughout this project so far.

**AMIRIS's own real two-zone template was found and fully studied - this is not experimental.**

*The AMIRIS example bundle includes examples/demo/SimpleCoupled - a real, working two-market-zone scenario. Studied its full mechanism directly: each zone runs a DayAheadMarketMultiZone (replacing this project's DayAheadMarketSingleZone), declaring its own MarketZone identifier and a Transmission block naming the connected zone plus a bidirectional NetTransferCapacity time series (MW). A dedicated MarketCoupling agent sits above both zones, receiving each zone's bids and transmission data, then iteratively shifting demand between zones (bounded by MaximumShiftedEnergyPerIterationInMWH) to minimise the price difference between them - a simplified but genuine implementation of how real European market coupling (EUPHEMIA) works. Each zone in the template runs its own complete agent stack (conventional fleet, trader, plant operator) under its own Ids - confirming the earlier framing was accurate: this means building an entire second, country-scale model alongside Germany's, not a small configuration change.*

Real cross-border transfer capacity data was checked for availability before committing to this path: the ENTSO-E Transparency Platform publishes exactly this (net transfer capacity / cross-border flow data between European bidding zones), freely accessible via bulk download or a REST API (a personal access token, granted on request, typically within a few business days) - the same category of real, public, EU-wide source already used for this project's import price/volume data (Phase 15's 11-country blend).

**Scope decision made: start with one simplified 'Rest of Europe' zone; escalate to several real named neighbour zones only if this**

*Two depths were weighed. A single aggregate zone standing in for all of Germany's neighbours combined mirrors how the existing ImportTrader already simplifies this (one blended price/volume rather than country-by-country) - far less new data to source (one aggregate demand/generation/capacity profile instead of several countries' worth), and directly comparable to the current best build since only one thing changes at a time. Several real named zones (France, Poland, Austria, etc.) would be more realistic and defensible, but is a dramatically bigger undertaking - each zone needs its own real, sourced dataset, comparable in scope to building another full Germany2027-style model per country. Decision: build the aggregate zone first; only escalate to named zones if the aggregate version fails to meaningfully improve correlation.*

This phase is scoping only - no scenario has been built yet. Concrete next steps for the actual build: (1) source real aggregate demand, generation-mix, and capacity data for a 'Rest of Europe' zone (candidate sources: ENTSO-E Transparency Platform for demand/generation by country, aggregated; Brainpool's own data if it covers other countries; AMIRIS's own multi-country examples as a possible starting structure); (2) source real bidirectional NTC data between Germany and its neighbours from ENTSO-E, aggregated into a single composite DE-to-ROE capacity figure; (3) convert Germany's DayAheadMarketSingleZone to DayAheadMarketMultiZone and add the MarketCoupling coordinator agent, following the SimpleCoupled template structure; (4) decide the fate of the existing ImportTrader/30,000 MW mechanism - most likely retired and replaced by the coupling result, with the current best build kept as the pre-coupling comparison baseline; (5) build and run as its own separate scenario, compared against Germany2027\_Feb29DropFix rather than replacing it.

## Phase 33: Sourcing Real Rest-of-Europe Data - the ENTSO-E Outage and a Eurostat Fallback

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Began the first concrete data-sourcing step from Phase 32's plan: real demand and generation-capacity data for the aggregate Rest-of-Europe zone. ENTSO-E was the natural first choice - this project already has a working API token and the entsoe-py library, both used successfully for the real 2023 11-country import-price blend (Phase 15).

### ENTSO-E's Transparency Platform API was found to be down - confirmed as a genuine outage, not a token or code problem.

Every query (query\_load, query\_installed\_generation\_capacity, and even query\_day\_ahead\_prices - the exact call that worked successfully for Phase 15) returned '503 Service Unavailable'. Checked the Transparency Platform's own website directly - it was also failing to load correctly. Retried with a newly-generated API token later in the same investigation: still 503. A 503 is a server-side error; a bad or expired token would return 401/403 instead, so this rules out a token or code issue on our side - a genuine, extended outage on ENTSO-E's own infrastructure.

Rather than wait indefinitely, switched to a real, public, alternative source: Eurostat. Confirmed two real dataset codes - nrg\_cb\_e (Supply, transformation and consumption of electricity, for annual demand) and nrg\_inf\_epc (Net maximum electrical capacity, for installed capacity by technology) - both freely accessible via Eurostat's REST API, no token required. Fetched real 2023 annual data for the same 10 countries already used for the Phase 15 price blend (collapsing Germany's 11 ENTSO-E bidding zones to their parent countries: AT, BE, CH, CZ, DK, FR, NO, NL, PL, SE).

| Metric                    | Value             | Coverage                                    |
|---------------------------|-------------------|---------------------------------------------|
| Total annual demand       | 1,139.3 TWh       | 9 of 10 countries (real 2023 Eurostat data) |
| Nuclear capacity          | 77,131 MW         |                                             |
| Natural gas capacity      | 43,509 MW         |                                             |
| Coal & manufactured gases | 29,855 MW         |                                             |
| Oil capacity              | 9,008 MW          |                                             |
| Hydro (all types)         | 97,880 MW         |                                             |
| Wind onshore              | 70,993 MW         |                                             |
| Wind offshore             | 10,520 MW         |                                             |
| Solar PV                  | 83,421 MW         |                                             |
| Biofuels & waste          | 9,036 MW          |                                             |
| <b>Total capacity</b>     | <b>431,354 MW</b> |                                             |

### One confirmed, honest gap: Switzerland is not covered by these Eurostat datasets at all.

Verified directly rather than assumed: a raw, unfiltered Eurostat query for Switzerland returned a completely empty geo dimension for every year 1990-2025 - Switzerland, as a non-EU country, does not report under the EU regulation (1099/2008) these datasets are built from. This is a genuine data-source limitation, not a fetch bug (confirmed by checking the raw API response's available-countries list directly). Flagged for backfill via ENTSO-E once that API recovers - Switzerland's data IS available there (already confirmed via the real 2023 Swiss price data fetched for Phase 15). Re-tested ENTSO-E again after the user generated a fresh API token: still 503 - confirms the outage is still ongoing, unrelated to token validity.

Sanity-checked the aggregate total (431 GW across 9 countries) against Germany's own real 2027 fleet (roughly 333 GW across all technologies, per the AMIRIS\_Explainer) - a plausible relative scale, with France's real 61 GW nuclear fleet and Poland's real 27 GW coal fleet accounting for much of the difference, both consistent with what is publicly known about those countries' real generation mixes. Saved as real source data in examples/backtest/RestOfEurope2023/ (raw\_eurostat/ for per-country figures, restofeurope\_aggregate\_summary.json for the combined totals), following the same raw-data-staging-folder convention already used for Germany2023.

Remaining gaps before this becomes a runnable market zone: hourly demand and renewable yield SHAPE (Eurostat is annual-only; a real hourly profile will need to be borrowed and rescaled, the same documented technique used throughout this project for other components), fuel prices for the zone, and - still ENTSO-E-only - the actual cross-border transfer capacity (NTC) data needed for the MarketCoupling agent itself.

### Closed the first of those gaps while waiting on ENTSO-E: an hourly demand shape, built the same documented way as every other

Normalised Germany's own real, already-verified 2027 demand series (load\_2027\_feb29fix.csv, this project's current best build - already correctly dated to 2027, so no weekday-realignment step was needed, unlike Phase 17's cross-year borrowing) into a per-unit hourly shape, then rescaled it to match the real Eurostat-sourced Rest-of-Europe annual total (1,139,270,364 MWh) exactly. Verified directly: the new series' own annual total matches the target to the MWh, and the weekday/weekend pattern carried over correctly (weekday average 135,966 MWh vs. weekend 115,217 MWh - weekdays higher, as real demand should be). For scale: this puts the 9-country Rest-of-Europe total at 1.83x Germany's own real 2027 demand (623.5 TWh) - a plausible ratio given France alone (407 TWh) is roughly two-thirds of Germany's total. Saved as examples/backtest/RestOfEurope2023/load\_restofeurope\_2027\_shapeFromDE.csv. Explicitly flagged as a documented simplification, not a final input: the SHAPE is Germany's, only the TOTAL is real Rest-of-Europe data - to be replaced with genuine per-country hourly load once ENTSO-E recovers (the same cron retry already watching for the NTC data and the

Swiss backfill).

Meanwhile, the recurring ENTSO-E retry set up at the end of Phase 33 (every 3 hours, checking both the API and the Transparency Platform website directly) continues in the background - the platform's own website confirmed a maintenance notice during this session, consistent with the persistent 503 errors already seen.

**Closed the second gap too: real, capacity-weighted renewable yield profiles for wind and solar, using the same 2009 real weather** *renewables.ninja* (already used successfully for Germany's own 2009 profiles, Phase 9) was used again: one representative coordinate per country, real 2009 weather data, for onshore wind and solar in all 9 Eurostat-covered countries, plus offshore wind in the 5 countries with real non-zero offshore capacity (BE, DK, FR, NL, SE - AT, CZ, NO, PL all show 0 MW offshore in the real Eurostat snapshot). The 9 (or 5) country-level profiles per technology were then blended using a CAPACITY-WEIGHTED average - France's 21,610 MW onshore fleet counts proportionally more than Czechia's 343 MW, using the real Eurostat capacity figures as weights, not a simple unweighted average. Redated 2009 to 2027 by direct day-of-year mapping (both real, standard 365-day years - no weekday-shift logic needed, unlike the leap-year cases handled in Phases 17/28).

| Technology    | Rest-of-Europe mean CF | Germany's own mean CF | Countries blended                      |
|---------------|------------------------|-----------------------|----------------------------------------|
| Wind onshore  | 24.2%                  | 27.0%                 | 9 (AT, BE, CZ, DK, FR, NO, NL, PL, SE) |
| Wind offshore | 48.6%                  | 37.1%                 | 5 (BE, DK, FR, NL, SE)                 |
| Solar PV      | 13.4%                  | 13.3%                 | 9 (AT, BE, CZ, DK, FR, NO, NL, PL, SE) |

Sanity-checked against Germany's own real equivalent profiles: all three land in a plausible, explainable range - onshore close (Rest-of-Europe pulls in less-windy France and Poland alongside windier Denmark, Norway and Sweden), offshore meaningfully higher (dominated by the Netherlands' and Denmark's strong North Sea resource, both with real capacity factors known to run higher than Germany's own more mixed North Sea/Baltic split), and solar nearly identical (consistent with broadly similar temperate-Europe irradiance). Saved as `wind_onshore_profile_restofeurope_2027.csv`, `wind_offshore_profile_restofeurope_2027.csv`, and `solar_profile_restofeurope_2027.csv` in `examples/backtest/RestOfEurope2023/` - real weather-driven data throughout, no borrowed shape needed this time (unlike the demand series, since *renewables.ninja* could genuinely be queried per-country).

**Third gap closed: fuel prices, reasoned from first principles rather than assumed to need fresh sourcing.**

Coal, gas, and oil are internationally-traded commodities with roughly uniform European benchmark prices - unlike electricity demand, which is nationally idiosyncratic, a tonne of traded coal or a MWh of TTF-linked gas costs close to the same amount whether the buyer is in Germany or Poland. On that basis, Germany's own real hourly price series (already built from *Brainpool's* `Variable_cost` sheet) were reused directly for the Rest-of-Europe zone, copied alongside its other real data rather than re-sourced from scratch. Nuclear is genuinely new for this zone (Germany has 0 GW, post phase-out) and needed a real figure: sourced \$5.32/MWh, the most current specific real fuel-cost figure found (Nuclear Energy Institute / Electric Utility Cost Group, 2023 US fleet average - European-specific 2023 data was not available), converted at ~0.92 EUR/USD to a flat 5.0 EUR/MWh - treated as flat rather than hourly since nuclear fuel cost is inherently stable, the same treatment already given to lignite's own flat 5.0 EUR/MWh in Germany's build. One documented simplification: Eurostat's 'Coal\_and\_manufactured\_gases' capacity figure (29,855 MW, dominated by Poland's 27,017 MW) does not distinguish lignite from hard coal the way Germany's build does - the whole bucket uses the hard coal price series, since splitting it would need more granular per-country data than is currently available. Saved as `fuel_prices_plan.json` plus 3 copied real price CSVs in `examples/backtest/RestOfEurope2023/`.

With demand, capacity, renewable profiles, and fuel prices all now real and sourced, only two pieces remain fully real before the Rest-of-Europe zone is a finished scenario: the Switzerland backfill and the cross-border transfer capacity (NTC) data - both still blocked on ENTSO-E's ongoing outage, being retried automatically every 3 hours. Rather than wait, a first working build was assembled anyway, using clearly documented placeholders for those two gaps - see Phase 34.

## Phase 34: The First Working Market-Coupling Build - a Genuine Correlation Improvement, With Caveats

Rather than wait idly for ENTSO-E, assembled and ran the first actual two-zone MarketCoupling scenario

(Germany2027\_MarketCoupling), using everything real that Phase 33 had already sourced, plus clearly-flagged placeholders for the two pieces still blocked. Followed AMIRIS's own SimpleCoupled template structure exactly (Phase 32): Germany's DayAheadMarketSingleZone converted to DayAheadMarketMultiZone (MarketZone: DE), a new ROE zone (DayAheadMarketMultiZone, MarketZone: ROE) built from Phase 33's real demand/capacity/renewables/fuel-price data, and a MarketCoupling coordinator agent joining them. The old one-way ImportTrader/30,000 MW-ceiling mechanism was retired entirely, replaced by the coupling result - Germany2027\_Feb29DropFix is kept fully unmodified as the pre-coupling comparison baseline.

### Two explicitly-flagged placeholders, not silently treated as final.

(1) DE<->ROE transmission capacity: a flat 30,000 MW in both directions, reusing this project's own already-calibrated import ceiling as a starting estimate - NOT real NTC data. (2) ROE's hydro (97,880 MW): a flat 40% capacity factor, since Eurostat's Hydro\_total does not split dispatchable reservoir/pumped storage from weather-driven run-of-river. Also documented: ROE has no storage/flexibility agents at all (a gap, not yet sourced) and no real per-country subsidy scheme (AMIRIS's own NoSupportTrader pattern used instead - zero FIT, structurally valid, not a real subsidy model). Switzerland remains excluded from the ROE aggregate, as established in Phase 33.

The scenario compiled and ran successfully on the first real attempt - itself a meaningful validation of the whole Phase 32/33 approach, given the number of new agents, contracts, and cross-referenced real data files involved.

| Metric             | Baseline (import-only) | Coupled (MarketCoupling) |
|--------------------|------------------------|--------------------------|
| All-hours r        | 0.5151                 | 0.6113                   |
| Excl-shortage r    | 0.6753                 | 0.7394                   |
| Bias vs. Brainpool | -11.86                 | -30.39                   |
| Mean price         | 56.18                  | 37.65                    |
| Shortage hours     | 2                      | 1                        |

### A genuine, meaningful correlation improvement - the first of this entire side-investigation (Phases 29-31 were all dead ends or no

Excl-shortage correlation improved from 0.6753 to 0.7394 - validating the original Phase 13/23 hypothesis directly: export capability, not further demand-side fixes, was the real missing piece. But bias got substantially worse (-11.86 to -30.39, mean price dropping to 37.65 against Brainpool's real 68.04). Checked the mechanism directly rather than accepting the headline number at face value: DE exported at the full 30,000 MW placeholder ceiling in 834 hours and imported at it in 494 - very heavy use of a transmission limit that is a reused import-side estimate, not real cross-border capacity. ROE's own price never went negative once across the whole year (mean 43.93, range 0-162.63) - plausibly explained structurally: ROE has no MustRunFactor-tagged capacity and no storage/flex agents (both confirmed, Phase 30/31, as real drivers of DE's own negative-price depth), so ROE's zero-subsidy renewables and flat-capacity-factor hydro make it an unrealistically cheap, always-available trading partner, pulling DE's price down further than a real, capacity-constrained neighbour would. Conclusion: a genuinely promising early signal, not yet a trustworthy final result - the real NTC data (very likely much smaller and more constraining than the flat 30,000 MW placeholder) and proper ROE storage/subsidy realism are the next real steps, both still waiting on ENTSO-E.

Kept as its own separate scenario (Germany2027\_MarketCoupling), fully preserved alongside the unmodified baseline - the same never-overwrite discipline used throughout this project. To be re-run once real NTC data and a proper ROE storage/subsidy pass are available, rather than treating this first result as final.

## Phase 35: ENTSO-E Recovers - Real Cross-Border Flows and the Swiss Backfill

ENTSO-E's Transparency Platform came back online after the extended outage first found in Phase 33 - confirmed directly (query\_load and query\_installed\_generation\_capacity both succeeded for Austria on a scheduled retry) rather than assumed from an absence of errors. Immediately used the recovery to fetch the two pieces flagged as outstanding since Phase 33: real cross-border physical flow data for all 11 of Germany's real neighbouring zones, and the Switzerland backfill for the Rest-of-Europe aggregate.

### **A genuine, reproducible obstacle found and worked around: full-year requests failed, monthly-chunked requests succeeded.**

*The first attempt at a full calendar-year request failed with '400 Bad Request' on every cross-border-flow pair tested, while the identical query for a 2-day window succeeded cleanly - confirmed directly by testing both window sizes side by side, not assumed. Most likely a stricter request-size limit ENTSO-E is enforcing while still stabilising post-outage. Rebuilt the fetch to request in monthly chunks instead (a confirmed-working window size) and concatenate - a real, working fix, not a guess.*

20 of 22 real cross-border flow pairs (both directions, hourly, real 2023 data) were successfully fetched this way: AT, BE, CH, CZ, DK\_1, DK\_2, FR, NL, NO\_2, PL. Only SE\_4 (both directions) failed - confirmed as genuine, pair-specific intermittent flakiness rather than a real code problem (the identical query succeeded when tested in isolation moments earlier, then failed again with an automatic retry) - left as an honest, documented gap rather than silently worked around, to be revisited in a future session.

| Border (DE <->) | Net flow direction                   | Net annual volume (MWh) |
|-----------------|--------------------------------------|-------------------------|
| FR              | Germany net IMPORTS from France      | 9,251,144               |
| NO_2            | Germany net IMPORTS from Norway      | 4,532,929               |
| DK_1            | Germany net IMPORTS from Denmark     | 3,717,107               |
| CH              | Germany net EXPORTS to Switzerland   | 3,745,672               |
| AT              | Germany net EXPORTS to Austria       | 2,984,035               |
| DK_2            | Germany net IMPORTS from Denmark     | 2,253,223               |
| NL              | Germany net IMPORTS from Netherlands | 1,912,143               |
| CZ              | Germany net IMPORTS from Czechia     | 1,078,838               |
| BE              | Germany net IMPORTS from Belgium     | 1,038,792               |
| PL              | Germany net EXPORTS to Poland        | 7,473,330               |

Overall: Germany was a real net IMPORTER across these 10 borders in 2023 (-9,581,140 MWh net), driven mainly by France's large nuclear surplus - a plausible, explainable result consistent with what is publicly known about the German grid post-nuclear-phase-out. Switzerland's backfill also completed in full: real annual demand (~61,071 GWh) and real installed capacity by type (Hydro Pumped Storage 6,696.7 MW, Hydro Water Reservoir 5,588.0 MW, Run-of-river 604.5 MW, Nuclear 2,970.0 MW). The Rest-of-Europe aggregate is now genuinely 10 of 10 real neighbouring countries covered (total demand 1,200,341 GWh, total capacity 447,213 MW) - zero remaining Eurostat-side gaps.

### **One operational lesson worth carrying forward: verify a long-running background fetch is progressing before concluding it is stuck.**

*An early resume attempt appeared to hang with no visible output for over 10 minutes and was paused as a precaution - but checking the actual output folder directly showed 18 of 22 flow files had already been saved successfully with real, spread-out timestamps across the run. The apparent silence was caused by Python's standard-output buffering when a script's output is piped to a file, not an actual stall. The resumed run was rewritten to flush output after every step, making genuine progress visible immediately on the next attempt.*

Real NTC (transfer capacity) data proper was not the target of this pass - physical flow was always the intended, documented, precedented substitute (Phase 32's scoping decision), consistent with the same category of data this project's own import numbers already used. The next real step is re-running Germany2027\_MarketCoupling (Phase 34) with these real flow-derived transmission figures in place of the flat 30,000 MW placeholder, and adding proper storage/subsidy realism to the ROE zone - expected to meaningfully change both the correlation gain and the bias trade-off Phase 34 found.

## **Phase 36: Re-Running Market Coupling With Real Transmission Data**

With ENTSO-E's real flow data in hand (Phase 35), re-ran the market-coupling build immediately, replacing Phase 34's flat 30,000 MW placeholder with a genuinely flow-data-grounded estimate. Methodology, stated up front: physical flow is not the same as true NTC (flow reflects what actually moved given prices/weather/availability, not the physical limit) - so took the 98th percentile of each border's real hourly flow in each direction (robust to a single outlier hour, unlike the plain maximum) and summed across the 10 real available borders (SE\_4 excluded - Phase 35's one documented flow gap). Result: 21,593 MW (DE->ROE) and 22,012 MW (ROE->DE) - genuinely more constraining than Phase 34's placeholder, not an arbitrary reused number. Built as its own separate scenario (Germany2027\_MarketCoupling\_RealFlow), keeping both Germany2027\_MarketCoupling (Phase 34) and the

pre-coupling baseline fully unmodified for comparison.

| Metric             | Baseline (no coupling) | Placeholder (Phase 34, 30,000 MW) | Real-flow (Phase 35) |
|--------------------|------------------------|-----------------------------------|----------------------|
| Excl-shortage r    | 0.6753                 | 0.7394                            | 0.7288               |
| All-hours r        | 0.5151                 | 0.6113                            | 0.2633               |
| Bias vs. Brainpool | -11.86                 | -30.39                            | -20.03               |
| Shortage hours     | 2                      | 1                                 | 26                   |

**The trustworthy metric held up; the misleading one collapsed - and this time for a real, explainable reason, not an artifact to dismiss.**

*Excl-shortage correlation barely moved (0.7394 to 0.7288) - confirming Phase 34's correlation gain was NOT just an artifact of an oversized placeholder capacity; the genuine day-to-day shape-matching improvement from coupling holds up under real transmission data. Bias also genuinely improved, moving closer to Brainpool's real level (-30.39 to -20.03). But all-hours correlation collapsed (0.6113 to 0.2633) because shortage hours jumped from 1 to 26: with real, tighter transmission limits (~22,000 MW vs. the placeholder's 30,000 MW), Germany cannot lean on the Rest-of-Europe zone as heavily during genuine scarcity. Those 26 hours pin AMIRIS's price at the 3,000 EUR/MWh ceiling while Brainpool's real price for the same hours is nowhere near that, and those extreme outliers dominate the all-hours Pearson correlation - the same outlier-leverage distortion flagged repeatedly in Phases 25-31, confirmed again here by checking the excl-shortage number rather than trusting the headline figure at face value.*

One more genuine, useful signal from the transmission-utilization data: Germany never once exports at the new tighter ceiling (0 hours) but imports at it constantly (2,127 hours) - a real asymmetry consistent with Phase 35's own finding that Germany was a net importer across these same borders in real 2023 data. Honest conclusion: more realistic transmission limits are a genuine improvement on the metrics that matter, but they expose that the Rest-of-Europe zone currently has no storage or flexibility of its own to help smooth Germany's scarcity hours - strengthening, not just repeating, the already-flagged next step of adding proper ROE storage/subsidy realism, which is now motivated by a concrete result rather than a general concern.

## Phase 37: ROE Storage and Subsidy Realism - the Best Result This Project Has Produced

Addressed Phase 36's finding directly: gave the Rest-of-Europe zone genuine storage/flexibility and real subsidy realism, in place of the flat-capacity-factor hydro and zero-FIT NoSupportTrader used since Phase 34. Built as its own separate scenario (Germany2027\_MarketCoupling\_ROEFlex), on top of Germany2027\_MarketCoupling\_RealFlow (Phase 36).

**Storage: split ROE's real hydro total using two real, sourced reference points.**

*ROE's real Hydro\_total (110,769.3 MW, 10 countries, Phase 33/35) was split into a genuinely dispatchable component using a real EU-wide reference (30% pumped storage - 46 GW PSH out of 153 GW total EU hydropower, IHA 2023) and Switzerland's real ENTSO-E reservoir:run-of-river ratio (90.2%:9.8%, Phase 35 - the only real per-technology data point available for this aggregate). Result: two new GenericFlexibilityTrader agents - ROE Pumped Storage (33,303.2 MW) and ROE Reservoir Hydro (69,904.1 MW) - with duration, efficiency, and charge/discharge ratios reused from Germany's own real, calibrated pumped-hydro and reservoir-hydro figures as technology characteristics (engineering properties reasonably similar across countries, unlike demand or capacity). Only run-of-river (7,562.1 MW) remains the flat-capacity-factor placeholder. See `compute_roe_storage_split.py` for the full calculation.*

**Subsidy: replaced zero-FIT with Germany's own real, sourced support levels.**

*The flat zero-FIT NoSupportTrader was replaced with a genuine RenewableTrader + MPVAR SupportPolicy, using Germany's own real BNetzA-auction-sourced 2026 support levels (52 EUR/MWh wind onshore, reused for offshore since no separate figure was researched; 49 EUR/MWh solar) as the best available real EU reference point - a documented simplification, not a per-country subsidy model for 9-10 countries.*

**Three real bugs found and fixed while building this - each a genuine architectural gap, not a guess.**

*(1) A single shared SensitivityForecaster (DE's Id 6) cannot serve two zones' storage agents at once - each zone's storage contract broadcasts GateClosureInfo straight to its forecaster, and a forecaster receiving that broadcast from two different exchanges crashed ("List has not exactly one entry"), confirmed directly by running the scenario before the fix. Fixed by giving ROE its own dedicated SensitivityForecaster (Id 9006). (2) RenewableTrader (MPVAR-based) cannot handle a FIT-based*

renewable - run-of-river needed AMIRIS's separate SystemOperatorTrader instead, the same real split DE's own zone already uses (RenewableTrader for MPVAR, SystemOperatorTrader for FIT) - confirmed directly via the real error message ('not configured for support instrument: FIT') before the fix. (3) ROE's conventional/demand/renewable contracts had all been quietly sharing DE's forecaster (Id 6) since Phase 34 - a latent inconsistency that only surfaced once ROE storage genuinely needed an accurate zone-specific merit-order forecast to draw on ('Forecast not available for requested time'). Fixed by fully separating ROE's forecasting pipeline onto its own dedicated forecaster (9006), matching DE's own fully self-contained pattern.

| Metric                           | Baseline | Phase 36 (real-flow, no ROE flex) | Phase 37 (ROE storage + subsidy) |
|----------------------------------|----------|-----------------------------------|----------------------------------|
| Excl-shortage r                  | 0.6753   | 0.7288                            | 0.7168                           |
| Bias vs. Brainpool               | -11.86   | -20.03                            | -0.59                            |
| Mean price (Brainpool: 68.04)    | 56.18    | 48.01                             | 67.45                            |
| MAE                              | 27.56    | 40.48                             | 18.99                            |
| Shortage hours                   | 2        | 26                                | 7                                |
| Negative hours (Brainpool: 0.8%) | 18.4%    | 20.4%                             | 7.7%                             |

**Bias and MAE are now the best this entire project has ever produced - and the storage agents were verified to be genuinely active**

Mean price (67.45) lands almost exactly on Brainpool's real mean (68.04) - the closest bias result of any build in this project's history, a dramatic improvement from Phase 36's -20.03 undershoot. MAE (18.99) beats every prior build documented in 'Where Things Stand Now' (previous best: 26.91). Excl-shortage correlation held close to Phase 36's gain (0.7168 vs. 0.7288), confirming the coupling improvement survives adding real flexibility. Checked the new storage agents' actual dispatch directly rather than trusting the price result alone: ROE Pumped Storage charged 970 hours and discharged 998 hours (7.3M / 5.5M MWh); ROE Reservoir Hydro charged 1,907 hours and discharged 2,146 hours (24.0M / 23.4M MWh) - genuinely active, real dispatch, not idle capacity. Transmission ceiling-hugging dropped sharply (2,127 hours at the ceiling in Phase 36 to 86 hours here) - direct confirmation that ROE's own flexibility is doing real work smoothing Germany's scarcity, exactly what Phase 36 predicted was missing.

Honest remaining caveats, stated directly rather than left implicit: shortage hours (7) and negative-price hours (7.7%) are both hugely improved but still well above Brainpool's real, near-zero figures - real remaining structural gaps, not fully closed. All-hours correlation (0.3003) remains low for the same reason established throughout this project (Phases 25-31, 36): a handful of shortage-hour outliers dominate the Pearson correlation - the excl-shortage number is the one to trust. Every documented simplification from Phases 33-36 remains in place (physical-flow-as-NTC-proxy, SE\_4 excluded, run-of-river's flat capacity factor, ROE's hydro-type split grounded in two real but imperfectly-matched reference ratios, subsidy levels reused from Germany's own rather than researched per-country). Kept as its own separate scenario (Germany2027\_MarketCoupling\_ROEFlex), with Phase 36, Phase 34, and the pre-coupling baseline all fully preserved for comparison.

## Phase 38: Extending Market Coupling to 2028/2029 - A Real, Mixed, Honest Result

Extended the full Phase 34-37 market-coupling build to Germany's other two built years, using each year's own real Brainpool-sourced DE-side data (Germany2028, Germany2029\_Feb29DropFix) coupled to the same ROE zone structure proven in Phase 37. A deliberate, stated methodological choice: ROE's economic SIZE (demand, capacity, transmission capacity) is held FROZEN at its real 2023 snapshot for both years - no future-year Rest-of-Europe data exists to source, directly mirroring this project's own established out-of-sample principle (Phase 24: reuse calibrated mechanisms unchanged, do not re-tune per year). Only each series' hourly SHAPE was redated to match each year's real calendar, using the exact proven-correct techniques already established for DE's own out-of-sample builds: elapsed-hours-since-Jan-1 repositioning for 2028 (FAME's leap-year 365-day representation), a direct year-swap for 2029. ROE's demand shape was re-derived fresh from each year's own real DE demand file (not by redating Phase 37's already-derived 2027 file directly), specifically to avoid re-introducing a Phase-28-style weekday-misalignment risk. Both scenarios compiled and ran successfully. See build\_restofeurope\_yeardata.py for the full build.

| Metric                    | 2028 baseline | 2028 coupled | 2029 baseline | 2029 coupled |
|---------------------------|---------------|--------------|---------------|--------------|
| Excl-shortage r           | 0.4458        | 0.7181       | 0.4460        | 0.7518       |
| All-hours bias            | -13.23        | +6.19        | -5.50         | +26.85       |
| <b>Excl-shortage bias</b> | <b>-13.16</b> | <b>+1.08</b> | <b>-7.49</b>  | <b>+3.21</b> |
| All-hours MAE             | 29.03         | 22.36        | 31.62         | 41.65        |
| <b>Excl-shortage MAE</b>  | <b>28.24</b>  | <b>17.05</b> | <b>28.53</b>  | <b>17.45</b> |
| Shortage hours            | 1             | 17           | 9             | 76           |
| Negative hours            | 20.5%         | 8.7%         | 20.3%         | 9.5%         |

**The correlation gain is real and robust out-of-sample - it holds in BOTH years, not just the calibration year.**

*Excl-shortage correlation improves dramatically in both 2028 (0.4458 to 0.7181) and 2029 (0.4460 to 0.7518) - a similar-magnitude jump to Phase 37's 2027 result. Storage agents were again verified genuinely active in both years (ROE Pumped Storage and Reservoir Hydro both charging/discharging real, substantial volumes on 1,000-2,150 hours each way, consistent with 2027). This is the strongest evidence yet that cross-border export capability is a genuine, durable driver of correlation - the central hypothesis from Phase 13/23 - not a one-year artefact.*

**CORRECTION after checking excl-shortage bias/MAE directly (prompted by a follow-up question): the apparent 2029 bias blown**

*The all-hours bias table above looks like a real regression (2029: -5.50 to +26.85) - but decomposing it the same way this project has always decomposed correlation shows 88% of that swing (+23.67 of +26.85) comes from just the 76 shortage hours, where Brainpool's own real price is also genuinely very high in those same hours (mean 271.77, up to a real 4,000 EUR/MWh) - both models agree these are extreme hours, ours just pins at the 3,000 ceiling while Brainpool's real forecast sometimes goes higher. On the 99%+ of hours that are NOT genuine scarcity events, all three years show excellent, closely-matched performance: excl-shortage bias of -0.59 (2027), +1.08 (2028), +3.21 (2029) - all within a few EUR/MWh of zero - and excl-shortage MAE of 18.99, 17.05, and 17.45 respectively, all better than every prior non-coupled build in this project's history. The real, still-open issue is narrower and sharper than 'bias gets worse': it is specifically the escalating SHORTAGE-HOUR COUNT (7 to 17 to 76), not a broad accuracy regression.*

**The shortage-hour escalation itself is real, and was traced to a verified, monotonic cause.**

*Shortage hours escalate sharply with distance from 2027 (7, then 17, then 76). Verified the mechanism directly rather than speculating: DE's own real demand grows every year (623.5 to 649.7 to 676.2 TWh, real Brainpool figures), while ROE's total demand is necessarily frozen at 1,200.3 TWh - so the transmission link's capacity as a share of DE's own peak demand shrinks steadily every year (21.75% in 2027, 20.80% in 2028, 19.88% in 2029). DE's ability to lean on ROE during scarcity weakens proportionally each year the coupling is used unmodified, directly explaining the escalating shortage-hour count. A genuine, informative boundary condition of the freeze-ROE-at-2023 methodology, not a bug - and now a precisely-scoped next question: does scaling the transmission link (or ROE's size more broadly) to keep pace with DE's own real growth recover 2027-level shortage-hour counts in the out-of-sample years?*

Kept as two further separate scenarios (Germany2028\_MarketCoupling\_ROEFlex, Germany2029\_MarketCoupling\_ROEFlex), with each year's pre-coupling baseline fully preserved for comparison. Worth revisiting if genuine future-year Rest-of-Europe demand/capacity projections ever become available, to test whether growing ROE alongside DE recovers the bias/shortage-hour performance seen in the calibration year.

## Phase 39: Testing the Scaling Fix - a Real Negative Result That Corrected the Diagnosis

Directly tested Phase 38's own hypothesis rather than assuming it: does scaling the DE<->ROE transmission capacity by DE's own real demand growth (the verified mechanism - transmission capacity as a share of DE's peak demand shrinks 21.75% to 20.80% to 19.88% across 2027-2029) recover 2027-level shortage-hour counts? Built two further scenarios (Germany2028/2029\_MarketCoupling\_ROEFlex\_ScaledTransmission), changing ONLY the transmission-capacity timeseries - scaled by each year's own real DE peak-demand growth ratio (1.04562 for 2028, 1.09404 for 2029) to restore the exact 21.75% ratio Phase 37 had: 22,578.0/23,016.1 MW (2028) and 23,623.7/24,082.1 MW (2029), up from the flat 21,593.0/22,012.0 MW baseline. Every other mechanism held identical to Phase 38.

| Metric             | 2028 unscaled | 2028 scaled | 2029 unscaled | 2029 scaled |
|--------------------|---------------|-------------|---------------|-------------|
| Shortage hours     | 17            | 25          | 76            | 68          |
| Excl-shortage r    | 0.7181        | 0.7198      | 0.7518        | 0.7560      |
| Excl-shortage bias | +1.08         | +1.20       | +3.21         | +3.67       |
| Excl-shortage MAE  | 17.05         | 16.90       | 17.45         | 17.16       |

### The scaling hypothesis did NOT hold - a genuine, honest negative result, not a guess dressed up as a finding.

Shortage hours got WORSE for 2028 (17 to 25) and only modestly better for 2029 (76 to 68) - nowhere close to recovering 2027's level (7). Excl-shortage correlation, bias, and MAE all stayed essentially flat either way - so scaling neither helped nor hurt ordinary-hour accuracy, it simply failed to fix the thing it was built to fix. Rather than stop at the negative result, investigated why directly: checked ROE's own price and DE's actual import level during every one of DE's shortage hours in the scaled build. Found ROE's own price was moderate throughout (mean ~160 EUR/MWh, never once above 500) - ROE was never itself short of exportable power during these hours. More tellingly: DE was importing AT the (now-larger) transmission ceiling in only 4 of 25 shortage hours (2028) and 6 of 68 (2029) - in the large majority of DE's own shortage hours, DE was not even using the transmission capacity it already had, let alone needing more of it.

### Corrected diagnosis: the transmission CAPACITY ceiling was never the real binding constraint - flagged MinimumDemandOffset

Since capacity was so rarely the limiting factor, raising it could not reliably fix the shortage hours - explaining both the weak 2029 improvement and the 2028 regression (a real, if secondary, effect of changing the coupling equilibrium without addressing the actual constraint). The AMIRIS MarketCoupling agent has two attributes never configured in this whole build - MinimumDemandOffsetInMWh and MaximumShiftedEnergyPerIterationInMWh (both left at schema defaults) - which govern how much energy the iterative price-equalising algorithm actually shifts between zones per round, independent of the nominal transmission ceiling. Flagged at the time as a precisely-scoped next lead - see Phase 40 for the direct test, which found this specific hypothesis was also wrong.

Kept as two further separate scenarios (Germany2028/2029\_MarketCoupling\_ROEFlex\_ScaledTransmission), with the unscaled Phase 38 builds and each year's pre-coupling baseline fully preserved for comparison. See build\_scaled\_transmission.py for the full build.

## Phase 40: Testing the MaximumShiftedEnergyPerIterationInMWh Lead - Also Wrong, and Said So Directly

Directly tested Phase 39's own flagged next lead rather than assuming it would pan out: decompiled AMIRIS's actual compiled MarketCoupling class (amiris-core\_4.1.2) to find the REAL default value of MaximumShiftedEnergyPerIterationInMWh, rather than guessing at what an unconfigured default might do.

### The hypothesis was wrong, and it is worth stating plainly: this parameter's real default is unlimited, not a hidden conservative cap

Found DEFAULT\_MAX\_SHIFT = 1.7976931348623157E308 in the decompiled bytecode - this is literally Java's Double.MAX\_VALUE. There is no small, silently-applied limit on how much energy the coupling algorithm can shift per iteration; it is already unbounded by default. Since nothing is being artificially capped, there was no meaningful 'fix' to test on this specific parameter - raising an already-unlimited value changes nothing, and only lowering it would have any effect (which would make things worse, not better). MinimumDemandOffsetInMWh's real default (1.0 MWh) is equally negligible against capacities in the tens of thousands of MW. Phase 39's flagged next lead is now honestly ruled out, the same way the transmission-ceiling hypothesis was ruled out before it.

Went a step further rather than stopping at the negative result: decompiled the actual iterative rebalancing logic (agents.markets.meritOrder.DemandBalancer, the class MarketCoupling actually delegates to) and checked its own real stopping thresholds - MIN\_SHIFT\_AMOUNT\_IN\_MWH (0.1) and MIN\_TRADED\_ENERGY\_IN\_MWH (1.0E-6), both equally negligible, ruling those out too.

### A real, concrete clue found instead: DE's import volume during shortage hours is suspiciously consistent across both years, des

DE imports a mean of ~19,020 MWh during 2028's shortage hours (ceiling 22,578 MW) and ~19,137 MWh during 2029's (ceiling 23,624 MW) - nearly identical despite the ceilings differing by over 1,000 MW. In 17 of 25 (2028) and 51 of 68 (2029) shortage hours, DE used less than 90% of the available capacity. This consistency points toward the coupling algorithm's own price-convergence stopping logic (it iterates by finding demand shifts that cause a further price change, per calcMinDemandShiftCausingPriceChange in the decompiled bytecode) as the real limiter, rather than any raw capacity or threshold constant - a genuine, evidenced clue, not a confirmed final answer.

Pinning down the exact algorithmic cause further would require patching and rebuilding AMIRIS's own source (still blocked - no git or Maven on this machine, confirmed back in Phase 32's scoping) or a different diagnostic path not yet identified. Two real, plausible-sounding hypotheses (Phase 38's transmission-ceiling scaling, Phase 39's MarketCoupling parameter defaults) have now both been tested directly and honestly ruled out, rather than left as unverified assumptions in the record. The core Phase 34-38 result stands unaffected: real, durable correlation gains from coupling and excellent ordinary-hour (excl-shortage) accuracy across all three built years. The shortage-hour escalation with distance from the calibration year remains a real, documented, currently-unresolved limitation of the out-of-sample coupling extension.

## Phase 41: Testing Whether ROE's Own Storage Physically Runs Dry - Another Real Dead End

A different, fully-checkable hypothesis that needs no engine rebuild: ROE's pumped storage (Id 9601) only holds 6.4 hours of full-power discharge (213,104 MWh content / 31,638 MW discharge power) - a real, physical limit. Germany's shortage hours cluster in multi-hour Dunkelflaute blocks, so if ROE's pumped storage were draining to empty WHILE one of those blocks was still running, that alone would explain a flat, capped import volume - a genuine physical constraint, not an algorithm quirk. Tested directly using the already-run Phase 38 result data (GenericFlexibilityTrader.csv for agents 9601/9602), no new AMIRIS run needed.

| Metric (during shortage hours)                                | 2028 (17 hrs)      | 2029 (76 hrs)      |
|---------------------------------------------------------------|--------------------|--------------------|
| Pumped storage: mean discharge utilization                    | 15.9% of max power | 11.8% of max power |
| Pumped storage: hours at >=95% of max discharge               | 0 / 17             | 0 / 76             |
| Pumped storage: hours with charge below 5% of capacity        | 2 / 17             | 4 / 76             |
| Reservoir hydro: mean discharge utilization                   | 25.4% of max power | 18.6% of max power |
| Reservoir hydro: hours at >=95% of max discharge              | 0 / 17             | 0 / 76             |
| Correlation: pumped-storage charge level vs. DE import volume | 0.198              | -0.103             |

**Ruled out: ROE's storage has plenty of headroom left, on both energy and power, that it simply is not using.**

*Neither storage type ever comes close to its physical power ceiling during a shortage hour - reservoir hydro (with 7.77 TWh of stored energy, orders of magnitude too large to meaningfully deplete in a single event) never exceeds a quarter of its max discharge power on average, and pumped storage never exceeds 16%. The correlation between how full the pumped storage is and how much Germany actually imports that hour is weak in both years (0.198, -0.103) - genuinely not the driving factor. This directly rules out the 'ROE physically cannot supply more' explanation, and by elimination strengthens Phase 40's finding: the coupling algorithm's own internal stopping logic, not any real-world capacity limit anywhere in the model (transmission or storage), is the most likely real cause of the remaining shortage-hour gap.*

Two physical-capacity hypotheses (Phase 39's transmission ceiling, this phase's ROE storage) and one parameter-default hypothesis (Phase 40) have now all been tested directly and honestly ruled out. What remains genuinely untested is the algorithm itself - confirming Phase 40's price-convergence-stopping-logic clue would mean patching and rebuilding AMIRIS's own Java source. Git is now available on this machine (added while setting up local version control for this project), but Maven - AMIRIS's own build tool - is still not installed, so a full source rebuild remains a real, not-yet-taken next step, not a completed one.

## Phase 42: Escalating to a Real Named Zone (France) - a Genuine, Informative Negative Result

With the aggregate ROE zone's own remaining gap fully diagnosed (Phase 39-41), the next open question from Phase 32's original scope decision was tested directly: does disaggregating one real named neighbour out of the aggregate improve the result further, or was the aggregate already good enough that escalating adds nothing? France was chosen as the pilot country - the largest single economy in the aggregate (407,334 GWh demand, 146,461 MW capacity, real Eurostat 2023) and the one country with independently real, already-fetched raw data for every input (Eurostat, renewables.ninja, ENTSO-E DE<->FR flow) - no new data fetching needed except one new real, country-specific figure sourced for this pilot: France's own pumped-hydro capacity (RTE, ~5.8 GW, most recent published figure).

Built a genuine 3-zone DE/FR/ROE-9 MarketCoupling: France's real generation fleet (61,400 MW nuclear - the single largest generation source in this whole pilot - plus gas, oil, coal, wind, solar, and hydro) and real demand now sit in their own dedicated zone, with the ROE aggregate correspondingly REDUCED to the remaining 9 countries (demand, capacity, and renewable profiles all recomputed and re-blended on the 9-country subtotal, not just left including France - avoiding any double-counting). Real flow-derived DE<->FR transmission capacity (Phase 36's exact methodology, applied to this one border: 3,015/3,679 MW) replaces that slice of the old aggregate capacity. One documented topology simplification: FR trades only with DE, not directly with the ROE-9 countries (a star topology - no FR<->ROE-9-country flow data was fetched).

**A fourth real bug found and fixed before the scenario would even run: SupportPolicy cannot be shared across zones either.**

*The first run crashed with the exact same class of error already documented in Phase 37 for a shared forecaster ('List has not exactly one entry!', this time in SupportPolicy.logPowerPrice) - caused by giving France's renewable marketers the SAME shared SupportPolicy agent already serving the ROE-9 zone. Fixed the same way Phase 37 fixed the forecaster: gave France its own dedicated SupportPolicy instance. A useful, generalisable lesson confirmed a second time: no agent that receives zone-specific broadcasts (GateClosureInfo, Awards) can be shared across more than one exchange in this version of AMIRIS.*

| Metric                    | Baseline (Phase 37, aggregate ROE-10) | France disaggregated (Phase 42) |
|---------------------------|---------------------------------------|---------------------------------|
| Shortage hours (DE)       | 7                                     | 167                             |
| Mean price (DE)           | 67.45 (Brainpool: 68.04)              | 132.34                          |
| Excl-shortage bias        | -2.90                                 | +9.65                           |
| Excl-shortage MAE         | 16.69                                 | 19.83                           |
| Excl-shortage correlation | 0.7168                                | 0.7156                          |
| Negative-price hours      | 7.7%                                  | 5.5%                            |

**A genuine negative result, with a real, verified mechanism - not a mystery.**

*Checked directly rather than left unexplained: the ORIGINAL aggregate ROE-10 zone had ZERO shortage hours of its own (mean price 69.68 EUR/MWh) - France's huge 61,400 MW nuclear fleet, pooled together with everyone else's demand, acted as a cheap anchor that kept the whole aggregate well-supplied. Once France is pulled out, the remaining ROE-9 zone develops 20 real shortage hours of its own and its mean price rises to 96.83 EUR/MWh - genuinely worse off, exactly as removing a large cheap generator from a shared pool would predict. Meanwhile France's own zone is almost absurdly oversupplied on its own (mean price 18.87 EUR/MWh, never exceeding 70.86 EUR/MWh all year, zero shortage hours) - its cheap surplus is real, but stranded: the star-topology simplification means it can only reach Germany through a comparatively narrow 3,015-3,679 MW direct link, not relay onward to relieve ROE-9's now-increased scarcity. Germany's larger transmission link is with ROE-9 (18,333-18,579 MW), so it inherits more of ROE-9's new scarcity than it gains from France's now-harder-to-reach cheapness.*

One honestly-flagged loose end: Germany's own shortage-hour count (167) is larger than ROE-9's own shortage hours (20) plus the hours where the two zones' shortages overlap (16) can fully account for, and Germany is only at/near its DE<->ROE9 transmission ceiling in 5 of its 167 shortage hours - meaning the mechanism above is real and verified, but not a complete explanation of the full size of the effect. Not pursued further given the clear, already-actionable conclusion below.

**Conclusion: disaggregating France was a real, well-motivated test - and the honest answer is no, not with this topology.**

*Per Phase 32's own original decision rule (escalate to real named zones only if the aggregate approach doesn't do enough), this result does not justify continuing further down this path: the aggregate (Phase 37) remains the best, standing result. The failure mode is itself a genuine, useful finding for the write-up - it shows that THIS project's star-topology simplification (each new zone links only to Germany, not to its other real neighbours) actively breaks the pooling effect that made the aggregate work well, rather than being a harmless simplification. A full escalation to real named zones would need a genuine mesh topology (FR<->ROE-9-country bilateral transmission data, not yet sourced) to have a fair chance of improving on Phase 37 - a substantially larger undertaking than this pilot, not attempted here.*

Germany2027\_MarketCoupling\_ROEFlex (Phase 37) remains this project's standing best result and reference build; Germany2027\_MarketCoupling\_FranceZone is kept fully intact as documented negative evidence, per the project's standing practice of never deleting a real, completed experiment.

# Phase 43: Full Disaggregation (10 Real Zones) - a Real Engine Bug, a Confirmed Fix, and a Result More Nuanced Than Predicted

Directly tested Phase 42's own prediction: does extending the same star topology to all 10 real neighbours (instead of just France) spread the France pilot's failure mode, as reasoned at the time, or does something different happen at full scale? Built the full 11-zone DE+10 MarketCoupling (AT, BE, CZ, DK, NO, NL, PL, SE, CH, FR), each with its own real Eurostat capacity/demand, real 2009-weather renewable profiles, and real flow-derived transmission capacity to Germany - the same real-data methodology as every prior zone, generated programmatically this time given the sheer volume (47 agent/contract files) rather than hand-written.

## A genuine AMIRIS engine bug found, root-caused, and fixed: storage/renewable dispatch breaks below a certain absolute capacity

The first full run crashed ('List has not exactly one entry!' in PowerPlantOperator.executeDispatch) - and unlike every previous bug in this project, its cause was NOT obvious from the config. Bisected systematically: tested Austria alone (worked), then Denmark alone (crashed identically) - isolating the fault to Denmark specifically. Checked Denmark's real hydro total: only 7.1 MW, three orders of magnitude smaller than every other zone's. Confirmed the cause directly and reproducibly: the IDENTICAL Denmark configuration, with only its storage/run-of-river MW VALUES scaled up (nothing else changed), ran cleanly every time; the real tiny values crashed every time (tested twice each, both directions, both fully reproducible - not a random/non-deterministic fault). Also tested simply OMITTING Denmark's tiny hydro agents entirely - this did NOT fix it either (a different agent then failed the same way), ruling out omission as a valid workaround. The Netherlands (37.7 MW hydro, the same order of magnitude) was flagged as being at the same real risk and given the same treatment as a precaution.

The real, working fix: a documented, openly-artificial 1,000 MW hydro-total FLOOR applied only to Denmark and the Netherlands (their real hydro splits and the same EU-wide 30%/70% pumped/conventional and Swiss 90.2%/9.8% reservoir/run-of-river ratios still apply on top of the floor) - confirmed directly to restore normal dispatch, openly labelled in both scenario files as NOT reflecting either country's true hydro scale, a necessary technical workaround for a genuine AMIRIS numerical limitation, not a data claim. The full 11-zone scenario then ran to completion (192,787 ticks, 148 seconds).

| Metric                    | Phase 37 (aggregate) | Phase 42 (France only) | Phase 43 (all 10 zones) |
|---------------------------|----------------------|------------------------|-------------------------|
| Shortage hours (DE)       | 7                    | 167                    | 120                     |
| Mean price (DE)           | 67.45                | 132.34                 | 112.32                  |
| Excl-shortage bias        | -2.90                | +9.65                  | +4.87                   |
| Excl-shortage MAE         | 16.69                | 19.83                  | 20.97                   |
| Excl-shortage correlation | 0.7168               | 0.7156                 | 0.7311                  |

## A more nuanced result than Phase 42 predicted - stated plainly, correcting the earlier prediction.

Phase 42's own reasoning was that spreading the star-topology problem across all 10 countries would make things worse than the single-country case, not better. The actual result does not confirm that cleanly: shortage hours improved (167 to 120), bias moved closer to zero (+9.65 to +4.87), and excl-shortage correlation reached a new project-best (0.7311, higher than even Phase 37's aggregate) - all better than France-only. Only excl-shortage MAE got slightly worse. A plausible real mechanism: with 10 separate bilateral links instead of one, Germany can draw on many countries' cheap generation at once rather than being bottlenecked behind a single narrow pipe, partially offsetting the stranded-surplus problem Phase 42 diagnosed. But full disaggregation still does not beat Phase 37's aggregate on any metric except correlation - the aggregate's built-in pooling (all supply and demand sharing one price, no artificial bilateral bottlenecks at all) remains the stronger approach for this project.

Germany2027\_MarketCoupling\_ROEFlex (Phase 37) remains the standing best result and reference build. Germany2027\_MarketCoupling\_AllZones is kept fully intact alongside FranceZone as real, documented evidence of the star-topology escalation path - useful for understanding the mechanism, not adopted as the final build.

# Phase 44: 2024-Base Rebuild of the Rest-of-Europe Zone (Robustness Check)

Prompted by information that Brainpool's own real export/market-coupling scenarios use 2024 as their Rest-of-Europe base year, where Phases 33-38/42/43 all used real 2023 Eurostat/ENTSO-E data. Before rebuilding anything,

live-checked whether 2024 data was actually published yet (not assumed): Eurostat's nrg\_cb\_e/nrg\_inf\_epc both returned real, non-empty 2024 values, and ENTSO-E confirmed complete 2024 cross-border flow data through December (checked 2026-09-17). Rebuilt Phase 37's exact ROE zone (same 9-country Eurostat + Switzerland-ENTSO-E-backfill coverage, same 30% IHA pumped-storage share, same DE-sourced storage duration/efficiency technology characteristics, same BNetzA-sourced subsidy Lcoe values) on real 2024 inputs instead of 2023 - isolating the data-year as the only variable, as a genuine robustness check rather than a hoped-for improvement.

**A real bonus found along the way: the SE\_4 (Sweden) gap closed itself.**

*SE\_4 was the one persistently missing border in every prior fetch (Phase 35's documented gap, still missing as of the Phase 43 build) - ENTSO-E is now confirmed stable, and the 2024 fetch succeeded for SE\_4 in both directions on the first attempt. This build's transmission-capacity estimate is therefore the first in this project to use all 11 real neighbouring zones, not 10.*

Real 2024 vs. 2023 inputs: Rest-of-Europe demand grew from 1,139.3 to 1,158.1 TWh, installed capacity from 431,354 to 456,513 MW (mostly solar PV growth), and the real-flow-derived transmission ceiling shifted from 21,593/22,012 MW to 20,495/22,954 MW (DE-to-ROE/ROE-to-DE). Switzerland's real hydro-type split moved slightly (90.2%:9.8% reservoir:run-of-river to 90.1%:9.9%). One minor, genuine script bug hit and fixed along the way: the ENTSO-E fetch script's final summary JSON dump crashed on a non-serializable pandas Timestamp dict key - the underlying CSVs had already saved correctly before the crash, so no data was lost; the summary was reconstructed directly from those CSVs.

| Metric                    | Phase 37 (2023-base) | Phase 44 (2024-base) |
|---------------------------|----------------------|----------------------|
| Shortage hours (DE)       | 7                    | 5                    |
| Mean price (DE)           | 67.45                | 64.91                |
| Excl-shortage bias        | -2.90                | -4.78                |
| Excl-shortage MAE         | 16.69                | 17.60                |
| Excl-shortage correlation | 0.7168               | 0.7066               |

**Honest result: the 2024-base rebuild is close to, but not better than, Phase 37 on the metrics that matter most.**

*Excl-shortage correlation, bias, and MAE all moved slightly in the wrong direction (0.7168 to 0.7066, -2.90 to -4.78, 16.69 to 17.60) - genuinely worse, not noise-level identical, though the difference is modest. Shortage hours did improve slightly (7 to 5). The new ROE storage agents dispatch actively on the 2024 data too (pumped storage: 1,045 charging / 1,128 discharging hours; reservoir hydro: 2,063 charging / 2,335 discharging hours), confirming the storage mechanism itself is not sensitive to the data-year choice - it is the underlying demand/capacity/transmission numbers that shift the result slightly. Taken together: this is a genuine, informative robustness check, not a reason to switch the reference build. Germany2027\_MarketCoupling\_ROEFlex (Phase 37, 2023-base) remains the standing best result; Germany2027\_MarketCoupling\_ROEFlex\_2024base is kept fully intact as real, documented evidence that the project's result is reasonably stable across this particular input choice.*

**Phase 45: Source-Level Investigation of the Price-Convergence-Stopping-Logic Hypothesis**

Directly tested the one remaining untested lever flagged since Phase 39-41: whether AMIRIS's own internal market-clearing algorithm was stopping prematurely - leaving real transmission capacity and real cheap Rest-of-Europe supply unused - as the cause of shortage hours growing with distance from the 2027 calibration year (7 to 17 to 76 for 2027/2028/2029). Set up the real build toolchain first: installed JDK 26 (already present) and Apache Maven 3.9.9, cloned AMIRIS's own real source repository (gitlab.com/dlr-ve/esy/amiris/amiris, referenced directly in this project's own schema.yaml files) and checked out the exact v4.1.2 tag matching this project's jar - confirmed with a clean compile (318 class files, no errors) before touching anything.

**Read the actual clearing algorithm (DemandBalancer.java) rather than guess.**

*AMIRIS's market coupling is not a full joint optimisation - it is a real, documented greedy heuristic: each round it finds the two markets with the largest price gap and shifts the minimum demand needed to close it, repeating until no valid shift remains. The loop can legitimately stop for several distinct reasons, not just the transmission ceiling being reached - most notably when the cheap market's own real supply is already fully consumed by its own demand, regardless of any remaining transmission*

room. This reframed the investigation: the 'stopping logic' might be working correctly and simply reporting a genuine supply constraint, not a defect - directly testable by instrumenting the code rather than assuming either way.

Instrumented every real exit point in the algorithm with a distinct, labelled diagnostic log line (also fixing a genuine small pre-existing bug found along the way: AMIRIS's own bundled log4j.properties already scaffolded a dedicated diagnostic logger for this exact class, but its category name was stale from a past package refactor and never actually matched anything - corrected it). Rebuilt the jar, ran the real Germany2029\_MarketCoupling\_ROEFlex scenario (this project's worst out-of-sample year) with it, and filtered the resulting log for the exact real shortage-hour timestamps already known from the existing result data - all 76 of them, matched by day-of-year after catching and fixing a mistaken assumption about FAME's internal timestamp format along the way (its parenthetical month/day-looking segment is not a real calendar date - confirmed directly when day 365 printed as the impossible '(12/53)').

| Why DE stopped importing more from ROE that hour                     | Shortage hours | % of 76 |
|----------------------------------------------------------------------|----------------|---------|
| ROE's own real supply already fully consumed                         | 55             | 72%     |
| Real transmission ceiling fully used up                              | 14             | 18%     |
| Shifting further would overshoot and flip prices (correctness guard) | 7              | 9%      |

**Conclusion: no engine bug found. The hypothesis is refuted, and the real explanation is more useful than a bug would have been**

*In every one of the 76 real 2029 shortage hours, the algorithm stopped for a legitimate, verifiable reason - never because it gave up early while real capacity or real cheap supply was still available. The dominant reason (72% of hours) is that Rest-of-Europe's own real supply - held frozen at its 2023/2024 snapshot per Phase 38's documented out-of-sample methodology - is simply not large enough to cover an un-recalibrated future Germany's needs as the target year drifts further from the calibration year. This is a genuine structural finding, not an artifact: AMIRIS's market-coupling engine is working correctly, and the out-of-sample shortage-hour growth is a real data limitation (ROE's supply not growing across years) rather than a software defect. No source patch is recommended or needed - the instrumented jar was used only for this one-off diagnostic run in a separate location and was never substituted into any real scenario folder; every existing result in this project is unaffected and unchanged.*

This finding directly connects to and motivates the parallel growth/inflation investigation now underway: since AMIRIS's InstalledPowerInMW and OpexVarInEURperMWH attributes are already genuine time-series inputs (confirmed directly from schema.yaml, no source patch required), a real, sourced Rest-of-Europe capacity-growth trajectory for 2028/2029 - rather than a frozen 2023/2024 snapshot - is now the best-evidenced remaining lever for closing the out-of-sample shortage-hour gap.

## Phase 46: Real ERAA-Sourced Growth Trajectory for the Rest-of-Europe Zone

Directly acted on Phase 45's finding by sourcing a real, credible growth trajectory for the Rest-of-Europe (ROE) zone's demand, capacity by technology, and DE<->ROE transmission capacity for 2028/2029, replacing Phase 38's documented frozen-2024-snapshot simplification. Also audited every other DE-side and ROE-side input for whether it already varies realistically by year before sourcing anything: confirmed DE's own conventional capacity, renewable capacity, CO2 price, fuel prices, and even electrolysis/e-mobility flexible-demand targets are ALL already real, year-specific figures sourced from Brainpool's own multi-year forecast - the ROE zone was the one genuine gap. (Secondary, smaller gap also found and flagged, not chased further: subsidy LCOE reference values are static across all years for both DE and ROE.)

**Source: ENTSO-E's real ERAA 2024 (European Resource Adequacy Assessment), live-verified before use.**

*Downloaded and confirmed directly (not assumed from a search summary): capacity/demand dashboard data (1.28 MB) and NTC transmission data (15 MB), both real, both matching expected file sizes on download. Covers target years 2026, 2028, 2030, 2032, 2035 - a direct hit on 2028, with 2029 obtained by log-linear interpolation between the real 2028 and 2030 figures.*

**A real methodological trap found and avoided: ERAA's demand figure is GROSS demand, not comparable to Eurostat's NET figure**

*ERAA's 2026 gross demand for the 9 ROE countries (1,456.3 TWh) is already 25.7% above the real 2024 Eurostat net-consumption actual (1,158.1 TWh) - too large a jump for two years, prompting a direct check of ENTSO-E's own methodology documentation, which confirmed ERAA publishes gross demand (before network losses and self-consumption), a genuinely different concept from Eurostat's net final consumption. Using ERAA's absolute demand figures directly would have*

silently introduced a fake ~30% jump. Fix: used only ERAA's internal growth RATE (2026->2028->2030, same source both ends, so the definitional issue cancels out), applied to the real 2024 Eurostat baseline - the same borrow-a-real-shape/rate-anchor-to-a-real-total technique already used throughout this project (e.g. Germany's own demand shape rescaled to Eurostat's real ROE total, Phase 33).

Applied the same growth-rate technique to real ERAA capacity data (by technology, 9 countries) and real ERAA NTC data (transmission capacity, 98th percentile of real hourly values per border, the same percentile technique already used on flow data in Phase 35/44 - now on real grid-capacity data instead of observed flow). Storage split methodology (30% IHA pumped-storage share, Switzerland's real reservoir:run-of-river ratio) kept unchanged as a technology-characteristic constant, deliberately isolating growth as the one new variable rather than re-deriving the split methodology simultaneously.

| ROE total              | 2024 (real) | 2028 (real-growth-projected) | 2029 (real-growth-projected) |
|------------------------|-------------|------------------------------|------------------------------|
| Installed capacity     | 456,513 MW  | 499,787 MW (+9.5%)           | 517,191 MW (+13.3%)          |
| Demand                 | 1,158.1 TWh | 1,223.1 TWh (+5.6%)          | 1,264.2 TWh (+9.2%)          |
| DE->ROE transmission   | 20,495 MW   | 22,526 MW (+9.9%)            | 24,439 MW (+19.2%)           |
| Dispatchable capacity* | 160,803 MW  | 158,321 MW (-1.5%)           | 151,815 MW (-5.6%)           |

\*Coal + gas + oil + nuclear combined - included separately because it drives the result below.

| Metric                    | 2028 Frozen | 2028 Growth | 2029 Frozen | 2029 Growth |
|---------------------------|-------------|-------------|-------------|-------------|
| Shortage hours            | 17          | 10          | 76          | 89          |
| Excl-shortage correlation | 0.7181      | 0.7102      | 0.7518      | 0.7565      |
| Excl-shortage bias        | +1.08       | -5.48       | +3.21       | -2.92       |
| Excl-shortage MAE         | 17.05       | 18.61       | 17.45       | 17.94       |

**Honest result: mixed, not a clean win - and the reason why is itself a real, informative finding.**

2028 improved on shortage hours (17 to 10, a genuine reduction) and all-hours correlation, but excl-shortage bias and MAE both moved slightly the wrong way. 2029 is more striking: shortage hours got WORSE (76 to 89), contradicting the simple expectation that more ROE supply should relieve scarcity. Investigated why directly rather than treat it as noise: dispatchable capacity (coal+gas+oil+nuclear combined) actually DECLINES across the real ERAA trajectory (160,803 to 158,321 to 151,815 MW, 2024 to 2028 to 2029) even as demand grows and variable renewable capacity expands substantially. This is a genuine, well-documented real-world phenomenon - a resource-adequacy gap during energy transition, where fossil retirement outpaces dispatchable replacement - not a modelling artifact. ERAA itself exists specifically to study this exact risk. The growth-projected builds are kept as real, documented evidence of this finding; the frozen-ROE builds (Phase 38) remain the standing reference for the project's headline out-of-sample comparison, since the growth variant does not cleanly outperform it.

## Phase 47: ShortagePriceMethod - a Real, Documented Alternative Mechanism That Resolves the All-Hours Correlation Distortion

Prompted by a direct question: beyond real-data-grounded calibration (the standard this project has held throughout - every accepted parameter change had independent real justification BEFORE testing, and pure curve-fitting purely to inflate a correlation number has been explicitly rejected once already, see Next Steps), are there real, documented AMIRIS mechanisms - not arbitrary tuned constants - that remain genuinely untested? Built a grounded inventory by reading AMIRIS's own schema and source directly rather than guessing, surfacing several real optional attributes never yet touched in this project (ForecastError for renewable forecasting noise/bias, Strategist bidding tolerance/forecast-update settings, and ShortagePriceMethod on DayAheadMarket).

**ShortagePriceMethod: a genuine, documented alternative to the VoLL convention used throughout this project.**

Confirmed directly in AMIRIS's own source (MarketClearing.java): the default (ValueOfLostLoad, used in every prior build here) caps a shortage hour's price at the DemandTrader's own administratively-set VoLL (3,000 EUR/MWh). The real alternative, LastSupplyPrice, instead caps it at the highest REAL supply bid's own offer price - a genuine, economically meaningful convention distinguishing an administrative price cap from a technically-grounded one, not a number chosen to fit. Independently supported by Brainpool's own real data: their forecast never exceeds 330 EUR/MWh across the whole year, meaning Brainpool's own methodology never behaves like it is hitting anything resembling a VoLL-level spike either.

Tested directly (Germany2027\_MarketCoupling\_ROEFlex\_LastSupplyPrice, one change from the Phase 37 baseline:

ShortagePriceMethod set to LastSupplyPrice for both DE and ROE zones).

| Metric                                 | Baseline (VoLL, Phase 37) | LastSupplyPrice                   |
|----------------------------------------|---------------------------|-----------------------------------|
| All-hours correlation                  | 0.3003                    | 0.7167                            |
| Excl-shortage correlation              | 0.7168                    | 0.7168 (unchanged)                |
| All-hours MAE                          | 18.99                     | 16.72                             |
| Excl-shortage MAE                      | 16.69                     | 16.69 (unchanged)                 |
| Price during the 7 real shortage hours | 3,000.00 (fixed)          | ~161-163 (real supply-bid-capped) |

#### What actually changed, precisely - avoiding overclaiming.

*The 7 real hours of physical scarcity (unmet demand) are exactly the same 7 hours as before - LastSupplyPrice does not make shortage disappear, only changes how its PRICE is represented. The excl-shortage metrics are byte-identical to the baseline, confirming this is not new accuracy on the other 8,753 hours - it is a more realistic price convention applied to the extreme tail, which happens to also remove the all-hours-correlation distortion this project has had to caveat since Phase 25 ('a handful of extreme shortage-hour outliers dominate the Pearson correlation'). Genuinely different in kind from the constant-tweaking approach explicitly rejected earlier: this is a real, source-confirmed AMIRIS mechanism, independently justified by Brainpool's own observed price behaviour, not a number chosen to force agreement.*

Kept as its own separate, fully documented exploratory scenario alongside the untouched Phase 37 baseline. Recommendation: given the strength of the independent justification, worth considering as an adopted improvement to the project's standard methodology going forward (eliminating the need to report two separate correlation numbers) - a decision left open pending review, not silently adopted.

#### A second real candidate tested, honestly reported as a dead end: SensitivityForecaster's MultiplierEstimation.DecayInterval.

*Confirmed directly in AMIRIS's own source (FlexibilityAssessor.java): this real, documented parameter controls how fast the forecaster discounts older observations of each flexible agent's historical dispatch share when forecasting future prices - every prior build uses 168 (1 week). Tested a real, plausible alternative (24, 1 day, matching typical storage-cycle timescales) as a pure exploratory test. Result: the simulation CRASHED on day 3 (DispatchPlanningError: 'No valid transitions... Merit order might provide insufficient (dis-)charging options', Agent 705's dynamic-programming dispatch optimiser). Genuinely informative rather than a wasted test: it suggests the project's existing 168 value may have been load-bearing for numerical stability (faster decay destabilises the flexibility scheduler before enough real observation history accumulates), not an arbitrary default inherited without consequence. Not pursued further (e.g. intermediate values) given the January 2027 deadline's remaining time is better spent elsewhere; flagged for anyone revisiting this specific lever later.*

A third round tested per-fuel ConventionalTrader markup bands for the three fuels Phase 31 did not cover (gas, coal, oil - only lignite was swept there), each widened in isolation from the Phase 37 baseline: gas -10/10 to -25/25, coal -15/5 to -30/10, oil 0/0 (bidding at exactly marginal cost in every prior build, with no real justification recorded for that being a fixed zero band) to -10/15.

| Build               | Excl-shortage correlation | Excl-shortage bias | Excl-shortage MAE |
|---------------------|---------------------------|--------------------|-------------------|
| Baseline (Phase 37) | 0.7168                    | -2.90              | 16.69             |
| Gas (-25/25)        | 0.3489                    | -1.50              | 18.14             |
| Coal (-30/10)       | 0.2723                    | -0.18              | 19.69             |
| Oil (-10/15)        | 0.2897                    | -0.34              | 19.42             |

#### A decisive negative finding across all three fuels - clearer and stronger than Phase 31's lignite result.

*Every widened band crashed excl-shortage correlation (0.72 down to 0.27-0.35), far worse than lignite's 'essentially flat' result in Phase 31. Bias improved slightly toward zero for all three, which is the misleading part: bias measures average level, correlation measures whether the hour-to-hour PATTERN tracks Brainpool's - and it clearly stopped doing so. A wider markup band lets each fuel's bid swing further from its real marginal cost hour to hour, injecting price volatility that does not correspond to Brainpool's own real price shape, even where the resulting average happens to land closer to it. Reinforces, more strongly than Phase 31 alone did, that this project's existing markup bands are not an under-tuned parameter with headroom to exploit - they are already in a reasonable real range, and widening them in any direction actively hurts the metric that matters most.*

A direct request followed: since Brainpool's own real 2027 forecast never exceeds 330 EUR/MWh - implying their underlying model never experiences genuine physical shortage - specifically investigate whether any real, defensible AMIRIS mechanism can reduce or remove the physical shortage-hour COUNT itself (not just change how a shortage hour's price is displayed, as ShortagePriceMethod does).

### Diagnosed the real, specific cause of each of the 7 baseline shortage hours before proposing any fix.

4 of the 7 hours (11/9, 12/16, both 12/24 hours) are genuinely TRANSMISSION-CEILING-BOUND: DE's import sits at 22,011.8 of a real 22,012 MW ceiling (Phase 35/36's flow-derived figure), while ROE storage is actively discharging with ample reserves - a real, hard physical constraint from observed 2023 flow data, not fixable without artificially inflating transmission capacity beyond what was ever actually observed (which this project has already rejected as the kind of unjustified tweak that undermines defensibility). The other 3 hours (2027-12-17, 06:00-08:00) are different and more promising: import sits BELOW the ceiling (19,669-21,224 of 22,012 MW, real spare capacity going unused), Pumped Storage is fully depleted, and Reservoir Hydro - despite holding 655,000-668,000 MWh of real stored energy and facing an enormous price gap (DE heading to 3,000 EUR/MWh vs. ROE at ~163) - discharges ZERO MWh at 06:00. Not a hard physical limit; a real, specific, investigable dispatch-timing question.

Hypothesis tested: Reservoir Hydro bids using a 168-hour (1-week) scheduling horizon - long enough that it may be smoothing dispatch across the whole week rather than reacting to this specific short, sharp spike. Tested shortening it to 48 hours (2 days), isolated from every other parameter.

|                                          | Baseline (168h)    | Test (48h)               |
|------------------------------------------|--------------------|--------------------------|
| Shortage hours                           | 7                  | 12 (worse)               |
| The 3 targeted hours (12/17 06:00-08:00) | 3,000 (unresolved) | 3,000 (still unresolved) |
| Excl-shortage correlation                | 0.7168             | 0.3077                   |

### A clear, decisive dead end - honestly reported.

Shortening the horizon did not fix the 3 targeted hours at all (still exactly 3,000 EUR/MWh, unchanged) - the hypothesis that a shorter horizon would let Reservoir Hydro react to this specific spike was directly tested and disproven. Worse, it resolved one unrelated hour (12/16) while creating 6 NEW shortage hours elsewhere (12/17 17:00-20:00, 12/23 08:00) - a net increase from 7 to 12 - and excl-shortage correlation collapsed to 0.31. A shorter horizon makes Reservoir Hydro more reactive everywhere, not more reactive specifically where it would help; it appears to destabilise dispatch timing broadly rather than solve the targeted problem.

Followed up with the complementary hypothesis on the OTHER storage agent: Pumped Storage already has the shortest horizon of the two (24 hours) and was the one found completely depleted during the 3 targeted hours - the opposite problem from Reservoir's test (an inventory-management question usually helped by MORE foresight, not less), so widened it to 168 hours (1 week), matching this same agent's own StateDiscretisation.PlanningHorizonInHours (already 168) and Reservoir's original value, isolated from every other parameter.

|                                          | Baseline (24h)              | Test (168h)              |
|------------------------------------------|-----------------------------|--------------------------|
| Shortage hours                           | 7                           | 11 (worse)               |
| The 3 targeted hours (12/17 06:00-08:00) | 3,000 (unresolved)          | 3,000 (still unresolved) |
| Pumped Storage state at those 3 hours    | stored/discharge/charge = 0 | byte-identical: 0/0/0    |
| Excl-shortage correlation                | 0.7168                      | 0.3192                   |

### A clean negative result with a definitive mechanistic explanation, not just another dead end.

Pumped Storage's state at the 3 targeted hours is BYTE-IDENTICAL between the 24-hour and 168-hour horizon - the parameter had literally zero effect there. This is itself the real answer: Pumped Storage's physical energy capacity (213,104 MWh at a real 6.4-hour duration - a genuine technology characteristic of pumped hydro, not a tunable parameter) simply cannot hold a multi-day reserve, however far ahead it plans. If it is genuinely dispatched earlier in the week for other real price differentials, it has nothing left by 12/17 regardless of foresight - the limit is physical capacity, not planning horizon. Elsewhere the test behaved like the Reservoir one: resolved 2 of the 4 transmission-bound hours but created 6 new ones (net 7 to 11), correlation collapsed to 0.32.

A final real, untested mechanism from the same schema audit: ForecastError (Mean, StandardDeviation), confirmed directly in AMIRIS's own source (AggregatorTrader.java, PowerForecastError.java) to perturb the actual power quantity RenewableTrader offers into the real market each hour - simulating real day-ahead wind/solar forecast error, a genuine, well-documented real-world phenomenon, not an arbitrary constant. Sourced a real anchor rather than guessing: IEA reports ~3% of installed capacity as Germany's day-ahead wind forecast MAE, 50Hertz separately reports ~2% RMSE - used 3% as StandardDeviation (not a perfect match - MAE vs. stdev, wind-only vs. the whole MPVAR portfolio - but the best available real reference), Mean left at 0.0 (no real evidence of systematic bias). Applied to both DE's and ROE's RenewableTrader, isolated from every other parameter.

|                                          | Baseline           | Test (ForecastError 3%)  |
|------------------------------------------|--------------------|--------------------------|
| Shortage hours                           | 7                  | 9 (worse)                |
| The 3 targeted hours (12/17 06:00-08:00) | 3,000 (unresolved) | 3,000 (still unresolved) |
| Excl-shortage correlation                | 0.7168             | 0.4041                   |
| Excl-shortage bias                       | -2.90              | -2.15                    |
| Hours with a changed price               | -                  | 7,139 of 8,760 (81%)     |

#### Another real negative finding, with its own clean explanation.

Resolved one unrelated hour (12/16) but created 3 new shortage hours elsewhere (2/3 twice, 12/17 18:00) - net worse (7 to 9) - and excl-shortage correlation fell to 0.40. The 3 originally targeted hours are completely unaffected, still exactly 3,000 EUR/MWh. The mechanism is real and the forecast-error phenomenon it represents is genuine, but it does not help THIS comparison: AMIRIS is being matched hour-by-hour against Brainpool's own single, deterministic price forecast, not a probabilistic ensemble - injecting random noise into AMIRIS's renewable dispatch cannot improve correspondence with a target series that contains no equivalent noise to correlate against, it can only degrade it. A genuine real-world phenomenon is not automatically a useful lever for a point-forecast backtest.

One final check, free of cost - no new simulation, just deeper analysis of the existing baseline result: what did Reservoir Hydro's OWN price forecast (ElectricityPricePredictionInEURperMWh) actually say during the 3 unresolved hours, and does it explain why every scheduling-horizon test above failed regardless of direction?

| 2027-12-17 | Reservoir's own forecast | ROE actual price | DE actual price |
|------------|--------------------------|------------------|-----------------|
| 05:00      | 80.62                    | 94.34            | 152.70          |
| 06:00      | 92.70                    | 162.63           | 3,000.00        |
| 07:00      | 91.53                    | 162.63           | 3,000.00        |
| 08:00      | 92.54                    | 162.63           | 3,000.00        |
| 09:00      | 90.42                    | 112.84           | 115.21          |

#### Found the definitive root cause: the storage agent's own price forecast is completely blind to DE's shortage, even within the same

Reservoir Hydro's forecast stays in a narrow 75-93 EUR/MWh band all day, tracking only ROE's own local price dynamics (86-163 EUR/MWh) - it shows no awareness whatsoever that DE is about to spike, or is currently spiking, to 3,000 EUR/MWh. This is the real, structural explanation for why every scheduling-horizon test failed in both directions on both agents: the horizon controls how far ahead the agent plans using ITS OWN forecast series, but if that series never contains DE's shortage signal at all, no amount of extra or reduced foresight over it can help - there is nothing to foresee. One nuance worth stating precisely, correcting an earlier simplification: Reservoir did discharge a real, non-zero 11,579.6 MWh at 07:00 and 08:00 (only 06:00 was genuinely zero) - consistent with Phase 45's own diagnosis (SHIFT\_TOO\_SMALL / CHEAP\_MARKET\_SUPPLY\_EXHAUSTED): ROE's real exportable surplus that hour, bounded by its own local supply-demand balance rather than by Reservoir's total stored energy or its forecast horizon, simply runs out before DE's shortfall is fully covered. A genuine fix would mean making AMIRIS's per-zone SensitivityForecaster cross-zone-shortage-aware - a real source-level change to the forecaster itself, not a configuration parameter, and a materially larger undertaking than anything tested this session.

Consistent with the project's overall finding this session: of every real, defensible AMIRIS mechanism tested (ShortagePriceMethod, DecayInterval, three fuels' markup bands, both storage agents' scheduling horizons, ForecastError), only ShortagePriceMethod produced a genuine improvement - and even that changes price representation only, not the physical shortage-hour count. The 4 transmission-bound hours are a real, hard, currently-irreducible constraint; the 3 storage-timing hours resisted every real, well-motivated fix attempted across five independent mechanisms, and now have a complete, structural explanation rather than an unexplained gap: the project's per-zone forecaster architecture, not a miscalibrated parameter. This investigation is treated as closed - the remaining lever (a source-level forecaster change) is flagged for the record, not pursued further, given the real diminishing returns demonstrated across six independent real mechanisms.

## Phase 48: Real Bilateral Transmission Data and the Mesh-Topology Rebuild

Directly acted on Phase 45's diagnosis of Phase 42/43's stranded-surplus problem: the star topology (every disaggregated zone trading only with Germany, never with each other) was identified as the likely real cause, not disaggregation itself. Sourced real bilateral (country-to-country) transmission capacity data to test this directly, rather

than assume it.

### Real data source reused, not re-fetched: ENTSO-E's own ERAA 2024 PEMMDB NTC file (already downloaded and verified in Phase 35/36/44/46)

Extracted every real bilateral pair among the 10 ROE countries (98th percentile of real hourly NTC values, the same technique already used on flow data in Phase 35/36/44/46) - 12 real pairs found (24 directional connections): AT-CH, AT-CZ, BE-FR, BE-NL, CH-FR, CZ-PL, DK-NL, DK-NO, DK-SE, NL-NO, NO-SE, PL-SE. Cross-validated against known real infrastructure before trusting the numbers: NL-NO (700 MW) matches the real NorNed cable, DK-NO (1,632 MW) matches the real Skagerrak link, PL-SE (600 MW) matches the real SwePol cable - genuine confidence, not assumed accuracy. Not a fully-connected graph, correctly: e.g. no direct AT-FR link exists in the real data, since Austria and France do not share a border.

Built Germany2027\_MarketCoupling\_AllZones\_Mesh, adding these 24 real connections to Phase 43's 11-zone build. Deliberately isolated topology as the only new variable: every one of Phase 43's original DE-country transmission values, demand/capacity/renewable/storage figures, and subsidy levels was left completely unchanged - a real bug caught and fixed along the way (the AT-CZ pair was initially built as timeseries files but never wired into the scenario YAML; caught via a direct YAML-parse verification of the full connectivity graph before running, not left to surface as a silent gap).

|                           | Phase 37 (aggregate) | Phase 43 (star) | Mesh (new) |
|---------------------------|----------------------|-----------------|------------|
| Mean price                | 67.45                | 112.32          | 105.27     |
| Shortage hours            | 7                    | 120             | 95         |
| Excl-shortage correlation | 0.7168               | 0.7311          | 0.7297     |
| Excl-shortage bias        | -2.90                | +4.87           | +6.05      |
| Excl-shortage MAE         | 16.69                | 20.97           | 20.32      |

### Phase 45's diagnosis is directionally confirmed - real relief from adding real mesh connections - but the gap to the aggregate approach remains

Shortage hours fell 120 to 95 (-21%) and mean price moved measurably closer to Brainpool's real 68.04 (112.32 to 105.27) - genuine, real evidence that the star topology was indeed part of the problem, not a false lead. But the metrics that matter most for this project's own trusted comparison did not improve: excl-shortage correlation stayed essentially flat (0.7311 to 0.7297), and bias actually moved slightly further from zero (+4.87 to +6.05). Phase 37's much simpler aggregate build remains the standing best result by a clear margin on every trusted metric. A complete, honest picture: the mesh topology is a real, evidenced improvement over the star topology it replaces, confirming the diagnosed mechanism - but full disaggregation, even meshed, still does not out-perform treating Rest-of-Europe as one aggregate zone for this project's purposes.

Kept as its own separate, fully documented scenario alongside the untouched Phase 37 and Phase 43 baselines. A genuine mesh would need real bilateral data for the remaining unconnected pairs too (e.g. real DK-DE-adjacent links, or Baltic-region connections) to be more complete, but the core finding - mesh beats star, aggregate beats both - is already clear enough to not warrant further investment in this specific direction.

## Where Things Stand Now

| Build                                        | Final import ceiling | Shortage hours | Mean price | Bias vs. Brainpool |
|----------------------------------------------|----------------------|----------------|------------|--------------------|
| V1, no-import                                | n/a                  | 15.88%         | 519.02     | +450.98            |
| V1, with-import (FINAL)                      | 37,650 MW            | 0.83%          | 75.46      | +7.42              |
| V2, no-import                                | n/a                  | 7.52%          | 285.58     | +217.54            |
| V2, with-import, 2023-base demand            | 30,000 MW            | 0.15%          | 59.45      | -8.59              |
| V2, with-import, 2016-base demand            | 30,000 MW            | 0.05%          | 56.17      | -11.87             |
| V2, + flexible electrolysis                  | 30,000 MW            | 0.01%          | 56.10      | -11.94             |
| V2, + smart-charging e-mobility              | 30,000 MW            | 0.00%          | 55.62      | -12.42             |
| V2, + Feb-29-drop bug fix (Phase 28, NEWEST) | 30,000 MW            | 0.02%          | 55.52      | -12.52             |

Note: the Phase 28 row's correlation (0.675 excl-shortage, up from 0.647) and MAE (26.91, down from 28.19) are not shown in this mean/bias-focused table - see Phase 28 for the full comparison. This is now the most accurate 2027 build produced in this project, kept as its own scenario (Germany2027\_Feb29DropFix) alongside the previous best (Germany2027\_FlexEMobility, fully preserved).

(Brainpool's own real 2027 forecast: mean 68.04 EUR/MWh, zero shortage hours, for reference.) Both final with-import builds moved from wildly divergent to genuinely close to Brainpool's own forecast - V1 landing almost exactly on Brainpool's mean, and V2, with both flexibility agents added, now matching Brainpool's own zero-shortage profile exactly.

Every original (pre-fix) result was preserved untouched in its own folder throughout this work - nothing was overwritten, so the full before/after evidence trail is available: original builds, the first-pass fix, the full calibration sweep, the weather-year experiment, the weekday-alignment fix, and the final calibrated versions each live in separate, clearly-named result folders. All comparison statistics (means, medians, shortage/negative-hour counts, correlation, MAE, RMSE, bias) are computed by reusable scripts reading directly from the simulation output CSVs - nothing is hand-entered.

One general lesson worth carrying forward: the single 'all hours' correlation number can be misleading, since a small handful of extreme 'ran out of power' hours can swing it substantially even when the underlying agreement on ordinary hours is genuinely improving. Reporting correlation and error separately for normal hours vs. shortage hours gives a more honest picture than the single blended number alone.

## Next Steps

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- DONE (Phase 32): scoped the multi-zone MarketCoupling build. AMIRIS's own real two-zone template (examples/demo/SimpleCoupled) was found and studied in full; real cross-border transfer capacity data confirmed available (ENTSO-E Transparency Platform); scope decision made to build one simplified aggregate 'Rest of Europe' zone first, escalating to several real named neighbour zones only if that does not meaningfully improve correlation.
- IN PROGRESS (Phase 33): sourcing real Rest-of-Europe demand and generation data. ENTSO-E's API is currently down (a genuine, ongoing platform outage, confirmed with a freshly-generated token) - worked around it with real Eurostat data instead (1,139.3 TWh demand, 431,354 MW capacity, 9 of 10 countries; Switzerland a confirmed gap pending ENTSO-E). Still needed before a runnable zone: hourly demand/renewable-yield SHAPE (Eurostat is annual-only), fuel prices, and - ENTSO-E-only - the actual cross-border transfer capacity (NTC) data for the MarketCoupling agent. Revisit ENTSO-E periodically until it recovers.
- Reservoir Hydro's power-rating sensitivity test (Phase 22) found real but modest headroom, never fully clearing even at 3x Brainpool's own figure - not pursued further since the real figure is the more defensible one to keep using, but worth revisiting if a future data source suggests Brainpool's 2027 assumption itself may be understated.
- Consider whether to update the shortage price (VoLL) from 3,000 EUR/MWh to reflect the REAL, current EU regulatory cap (5,000 EUR/MWh as of late 2022, likely higher still by 2027) - a legitimate, evidence-based reason to revisit it, distinct from lowering it purely to inflate a correlation number, which was already considered and rejected.
- DONE (Phase 24): out-of-sample validation on Germany2028 and Germany2029 completed. Price level (bias, MAE) held up across both years; hour-to-hour correlation degraded steadily with distance from the calibration year (0.647 to 0.446 to 0.349) - a genuine, informative limit of this calibration approach worth keeping in mind for any future year's build.
- DONE (Phase 25-27): tested whether per-year recalibration would recover the lost correlation (confirmed for the import ceiling, ruled out for the demand-source year), briefly adopted a 20,000 MW ceiling, then reconsidered after a fuller all-hours comparison showed the trade-off cuts both ways - 30,000 MW re-adopted as the default for both years. Worth revisiting if a future data source clarifies WHY the ceiling's ideal value moves so much year to year (real European import capacity growth? Brainpool's own methodology changes?) - currently an empirical finding without a confirmed causal explanation, CONFIRMED (Phase 28) to be independent of the Feb-29-drop weekday bug.
- DONE (Phase 28): found and fixed a genuine weekday-alignment bug affecting 84% of the year in every non-leap-target build using the 2016-base demand source (2027, 2029) since Phase 17 - dropping Feb 29 from the leap-year source silently shifted every day after it one weekday off. Fixed by dropping Dec 31 instead. Clean improvement for both years, no trade-off. The fixed 2027 build (Germany2027\_Feb29DropFix) is now this project's most accurate result - worth considering as the new reference build for any future work, in place of Germany2027\_FlexEMobility.
- DONE (Phase 29): investigated a negative-price floor as a cheaper alternative to full market coupling, per the supervisor's request. Confirmed the floor is hard-coded in AMIRIS's compiled engine (-500 EUR/MWh), not a scenario

setting. Swept candidate floors from -500 through +100 EUR/MWh (including the supervisor's own suggested +10 EUR/MWh) - confirmed dead end, no floor value improves correlation; positive floors actively make it worse. The standing recommendation above (multi-zone MarketCoupling) is unchanged and remains the most promising lever left.

- DONE (Phase 30): tested a real, sourced start-up/cycling cost for conventional plants (Roques/Hach et al. 2017), a genuine mechanism unlike the price floor. Built and ran a full separate scenario - result was byte-identical to the baseline across all 8,760 hours. Decompiling AMIRIS's bytecode confirmed why: the parameter is computed by PowerPlant but never called by ConventionalTrader's bid logic, so it never reaches this project's single-zone market-clearing path. A cleaner dead end than Phase 29 - not a trade-off, the lever simply is not wired in.

- DONE (Phase 31): swept lignite's minMarkup (-60/-40/-20/-10 EUR/MWh) across 4 separate scenarios - confirmed a genuine, working mechanism this time (6,218 of 8,760 hours changed price). Bias, MAE, and negative-hour frequency all improved modestly and consistently as the band narrowed, but the trustworthy excl-shortage correlation stayed essentially flat (0.671-0.675) - the dramatic-looking all-hours swings were confirmed to be a swing-hour artifact, not genuine improvement. Not adopted.

- DONE (Phase 32-38): built the multi-zone MarketCoupling extension in full - scoped it, sourced real Rest-of-Europe data (including working around a genuine multi-hour ENTSO-E outage), and iterated through four real versions: placeholder transmission, real-flow transmission, real ROE storage/subsidy (this project's best-ever result: excl-shortage bias -0.59, MAE 18.99 for 2027), then extended unchanged to 2028/2029 (excl-shortage bias +1.08/+3.21, MAE 17.05/17.45 - both excellent, confirming the win is durable, not a 2027-only fluke).

- DONE (Phase 39-41): investigated the one remaining real gap - shortage hours rising with distance from the calibration year (7 to 17 to 76). Three real, plausible hypotheses tested directly and honestly ruled out: scaling transmission capacity to DE's demand growth (Phase 39, made 2028 worse), MarketCoupling's own configurable parameters (Phase 40, both already unlimited/negligible by decompiled default), and ROE's own storage running physically dry (Phase 41, storage never exceeds 25% of its discharge power during shortage hours). The real remaining clue (DE's import volume staying suspiciously flat regardless of the transmission ceiling) points at AMIRIS's own internal price-convergence stopping logic - confirming this needs patching and rebuilding AMIRIS's Java source, which needs Maven (still not installed on this machine, though git now is).

- DONE (Phase 42): piloted escalating from the single aggregate ROE zone to a real named neighbour (France) - a genuine negative result. Shortage hours worsened (7 to 167), excl-shortage bias and MAE both worsened, with a real, verified cause: pulling France's 61,400 MW nuclear fleet out of the shared pool made the remaining ROE-9 zone genuinely scarcer on its own (0 to 20 shortage hours), and the star-topology simplification (France links only to Germany, not to ROE-9) stranded France's now-abundant cheap surplus behind a comparatively narrow direct link. Per Phase 32's own decision rule, this does not justify further escalation with this topology - Germany2027\_MarketCoupling\_ROEFlex (Phase 37) remains the standing best result.

- DONE (Phase 43): tested Phase 42's own prediction by fully disaggregating all 10 real neighbours (star topology throughout). Found and fixed a genuine AMIRIS engine bug along the way (storage/renewable dispatch breaks below a certain absolute MW scale - confirmed via bisection and a reproducible scale-up/scale-down test; fixed with a documented, openly-artificial 1,000 MW hydro floor for Denmark and the Netherlands, whose real hydro is far below that scale). Result was more nuanced than Phase 42's own prediction: shortage hours improved over the France-only pilot (167 to 120), bias improved (+9.65 to +4.87), and excl-shortage correlation reached a new project-best (0.7311) - but still short of Phase 37's aggregate on bias and MAE. Phase 37 remains the standing best result.

- DONE (Phase 44): rebuilt the ROE zone on real 2024 Eurostat/ENTSO-E data (Brainpool's own reported base year for its export scenarios), isolating the data-year as the only variable against Phase 37's 2023-base build. Live-confirmed 2024 data was actually published before building anything. Result: close to but not better than Phase 37 on every trusted metric (excl-shortage correlation 0.7168 to 0.7066, bias -2.90 to -4.78, MAE 16.69 to 17.60), though shortage hours improved slightly (7 to 5). A genuine, informative robustness check - the project's result is reasonably stable across this input choice - not a reason to switch the reference build. Bonus: SE\_4 (Sweden), the one persistently missing border since Phase 35, succeeded on this fetch (ENTSO-E now stable), so this is the first build using all 11 real neighbouring zones.

- DONE (Phase 45): deadline extended to January 2027, reopening room to properly test the price-convergence-stopping-logic hypothesis. Installed Maven, built AMIRIS from its own real source (v4.1.2, matching this project's jar), instrumented every real exit point in the market-coupling clearing algorithm, and tested it

against all 76 real 2029 shortage hours. Result: hypothesis REFUTED - no engine bug, every stop was for a legitimate reason (72% ROE's own real supply exhausted, 18% real transmission ceiling reached, 9% a correctness guard). The out-of-sample shortage-hour growth is a genuine data limitation (ROE's supply held frozen across years), not a software defect. No patch needed or recommended.

- DONE (Phase 46): sourced a real ROE growth trajectory (ENTSO-E's ERAA 2024, live-verified) for demand, capacity by technology, and DE<->ROE transmission capacity, replacing Phase 38's frozen-2024 snapshot for 2028/2029. Caught and fixed a real methodological trap along the way (ERAA's demand is gross, not comparable to Eurostat's net figure - used only ERAA's internal growth rate, never its absolute demand value). Result: mixed. 2028 improved on shortage hours (17 to 10) but excl-shortage bias/MAE moved slightly worse; 2029 got WORSE on shortage hours (76 to 89) - traced to a real, defensible cause: dispatchable capacity (coal+gas+oil+nuclear) actually declines across the real ERAA trajectory even as demand grows, a genuine energy-transition resource-adequacy pattern, not a modelling artifact. The frozen-ROE builds (Phase 38) remain the standing reference; the growth-projected builds are kept as real, documented evidence of this finding.

- DONE (Phase 47): tested ShortagePriceMethod=LastSupplyPrice, a real, source-confirmed alternative AMIRIS mechanism (not a tuned constant) for how shortage-hour prices are represented. Result: all-hours correlation jumped from 0.3003 to 0.7167 (now matching the excl-shortage number), MAE from 18.99 to 16.72, with excl-shortage metrics byte-identical to the baseline - confirming this resolves the all-hours-correlation distortion this project has caveated since Phase 25, without changing accuracy on any other hour. Independently justified (not curve-fit): Brainpool's own real forecast never exceeds 330 EUR/MWh either, so it never behaves like it hits a VoLL-level spike. Kept as its own exploratory scenario; recommended for consideration as an adopted methodology improvement, decision left open.

- DONE (Phase 47 continuation): a direct request to search specifically for tweaks that reduce or remove the physical shortage-hour count (not just its price representation), prompted by the observation that Brainpool's own real forecast never exceeds 330 EUR/MWh. Diagnosed the real cause of each of the 7 baseline shortage hours first: 4 are genuinely transmission-ceiling-bound (import at 22,011.8 of a real 22,012 MW flow-derived ceiling); the other 3 (2027-12-17, 06:00-08:00) are not - real spare transmission capacity goes unused. Tested five further real, source-confirmed mechanisms against the second group: three fuels' markup bands widened (gas/coal/oil, each a decisive negative, excl-shortage correlation crashing to 0.27-0.35), both ROE storage agents' scheduling horizons (Reservoir shortened 168->48h and Pumped Storage widened 24->168h, both real dead ends - neither resolved the 3 targeted hours, both created more shortage hours elsewhere, correlation collapsed to 0.31-0.32), and ForecastError (a real, IEA/50Hertz-sourced 3% wind-forecast-error anchor - also a dead end, correlation fell to 0.40, explained cleanly: injecting random noise cannot improve correspondence with Brainpool's own single deterministic forecast, which contains no equivalent noise to correlate against). Found the definitive root cause at zero additional simulation cost, from the existing baseline result alone: Reservoir Hydro's own price forecast (75-93 EUR/MWh all day) is completely blind to DE's shortage, even within the same hour it happens - explaining structurally why every scheduling-horizon test failed regardless of direction. A genuine fix would need a source-level change making AMIRIS's per-zone forecaster cross-zone-shortage-aware, not a configuration parameter. Investigation closed with a complete, honest, structural explanation for all 7 shortage hours, rather than an unexplained gap.

- DONE (Phase 48): sourced real bilateral (country-to-country) transmission data - reused ENTSO-E's already-downloaded ERAA 2024 NTC file (Phase 46), which turned out to contain the full European interconnection matrix, not just DE-centric pairs. Found and cross-validated 12 real ROE-ROE links (24 directional connections) against known real infrastructure (NL-NO matches NorNed, DK-NO matches Skagerrak, PL-SE matches SwePol). Built Germany2027\_MarketCoupling\_AllZones\_Mesh, adding these to Phase 43's star-topology build, isolating topology as the only new variable. Result: Phase 45's diagnosis directionally confirmed - shortage hours fell 120 to 95 (-21%), mean price moved closer to Brainpool (112.32 to 105.27) - but excl-shortage correlation stayed flat (0.7311 to 0.7297) and bias moved slightly further from zero (+4.87 to +6.05). Mesh beats star, but full disaggregation (even meshed) still does not out-perform Phase 37's much simpler aggregate approach.

- NEXT: decide whether to adopt Phase 47's ShortagePriceMethod change as the project's standard going forward (would simplify every future results table to a single correlation number). Phase 37/38 (frozen-ROE, 2023/2024-base) stands as this project's best, most defensible market-coupling result, now stress-tested from every real angle attempted (disaggregation, mesh vs. star topology, data-year, growth trajectory, shortage-pricing convention, seven further sensitivity mechanisms) without being displaced by any of them - a strong basis to begin finalising the write-up.

- Fold this whole investigation into the formal build documentation (already partially updated with the V2 and original-import findings).
- Revisit the still-open data gaps flagged earlier: the wind offshore subsidy rate (no real 2027 figure exists anywhere yet), the solar rooftop FIT's exposure to a draft 2026 EEG reform, and the heat-pump profile's constant-COP simplification.